

>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

**COMPREHENSIVE NONLINEAR
SEISMIC ASSESSMENT OF STEEL A
BEAM ELEMENT: AN OPENSEES
FRAMEWORK FOR STATIC
PUSHOVER, CYCLIC DEGRADATION,
STATIC TIME-HISTORY AND
DYNAMIC TIME-HISTORY ANALYSIS**

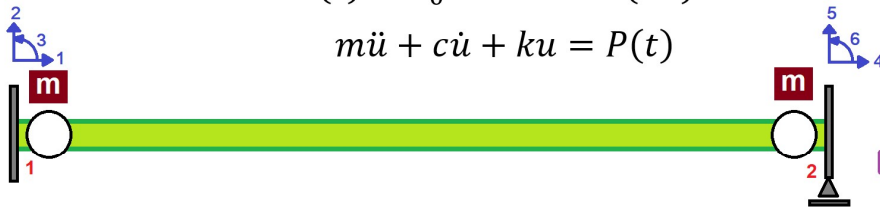
THIS PROGRAM IS WRITTEN BY SALAR DELAVAR GHASHGHAEE (QASHQAI)

$$\text{Axial Ductility Damage Index} = \frac{\varepsilon_d - \varepsilon_y}{\varepsilon_u - \varepsilon_y}$$

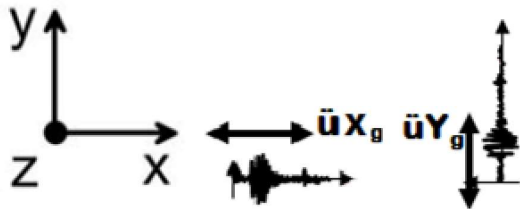
$$\text{Structure Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

$$P(t) = P_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$

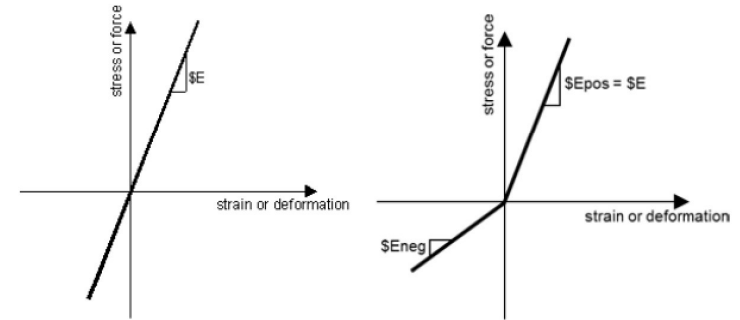
$$m\ddot{u} + c\dot{u} + ku = P(t)$$



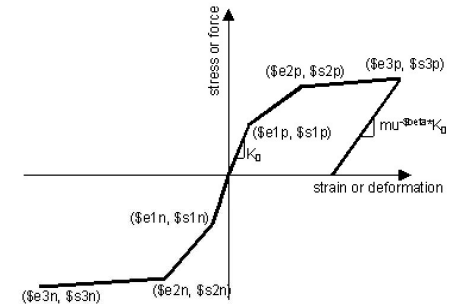
Length



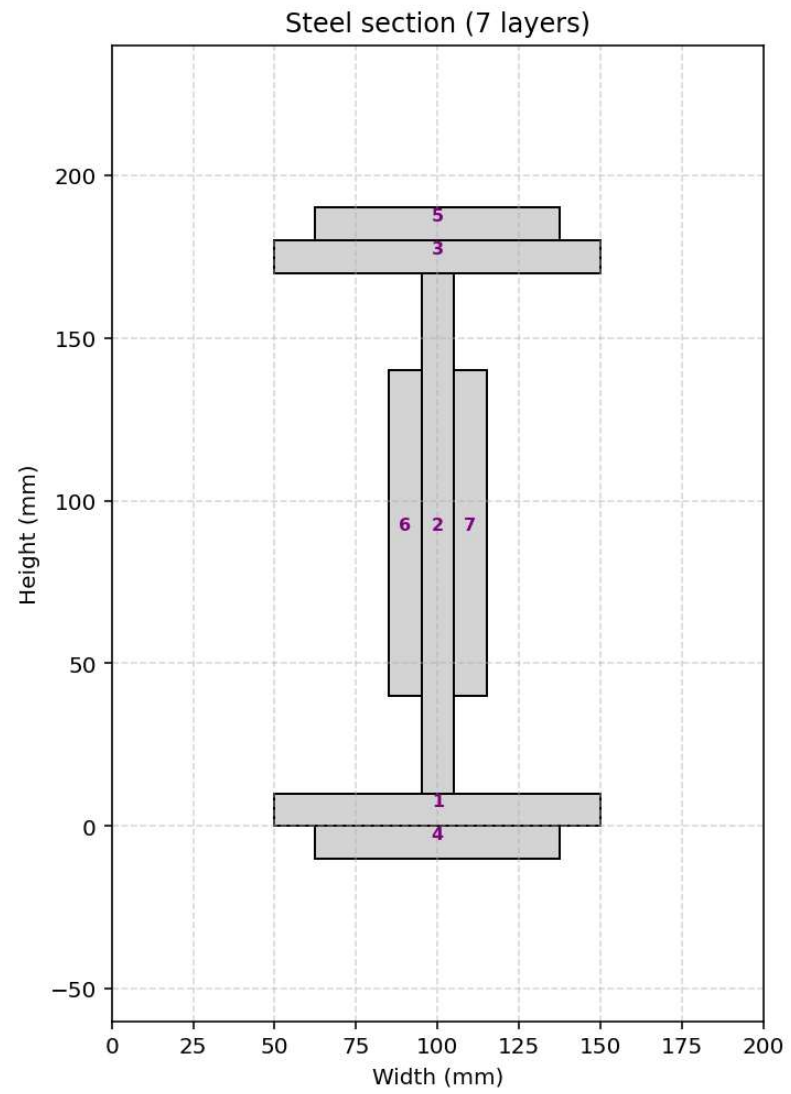
$$\rho A \frac{\partial^2 v}{\partial t^2} = P(t)$$



ELEMENT ELASTIC MATERIAL



ELEMENT INELASTIC MATERIAL



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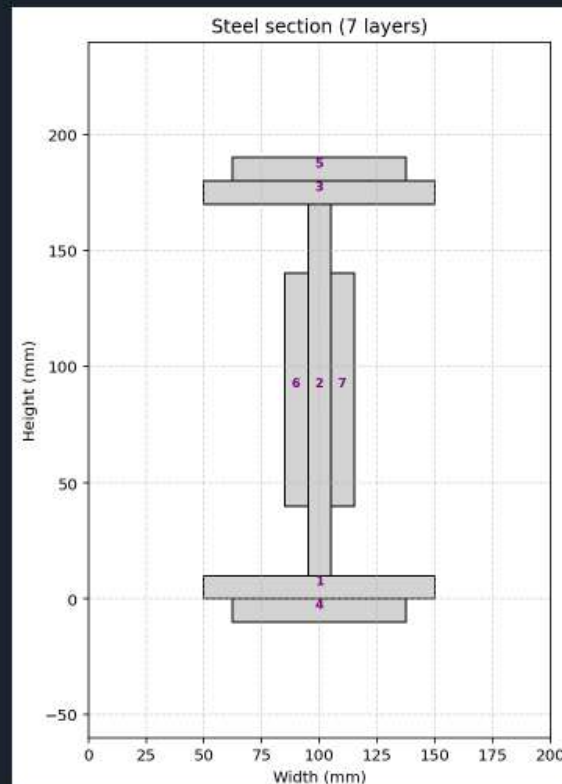
P(t)_BEAM_ONE_ELEMENT_RELEASE_M3.py X

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1 #####
2 # >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
3 # COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF STEEL A BEAM ELEMENT : AN OPENSEES FRAMEWORK
4 # STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY ANALYSIS
5 #
6 # EQUIVALENT VISCOUS DAMPING RATIO:  $\xi_{eq} = 100 * E_d / (4 * \pi * E_s)$ 
7 #
8 # ENERGY DISSIPATION CAPACITY INDEX =  $100 * E_d(\text{earthquake}) / E_d(\text{cyclic displacement})$ 
9 #
10 #  $P(t) = P_0 \sin(\omega t)$ 
11 #  $P(t) = P_0 \exp(-0.05\omega t) \sin(\omega t)$ 
12 #
13 # ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AND SYSTEMS AND EVALUATION
14 # OF ENERGY DISSIPATION CAPACITY INDEX
15 #
16 # THIS PYTHON SCRIPT IS WRITTEN BY SALAR DELAVAR GHASHGHAEE (QASHQAI)
17 # EMAIL: salar.d.ghashghaei@gmail.com
18 #####
19 """
20 Nonlinear Seismic Performance Assessment of a Beam Element:
21 An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Static, Cyclic, and Earthquake
22
23 This OpenSeesPy script performs rigorous nonlinear static and dynamic analysis of a Beam Element
24 for performance-based earthquake engineering. The 2D model incorporates both material nonlinearity
25 (elastic-perfectly plastic or hysteretic steel with strain hardening)
26 and geometric nonlinearity to capture P-delta effects
27 and large displacements-essential for collapse assessment.
28
29 Eight analysis protocols are implemented:
30 (1) [PERIOD] : Structural Period
31 (2) [STATIC] : Gravity load analysis establishing dead load state
32 (3) [PUSHOVER] : Displacement-controlled pushover generating full capacity curves
33 and plastic mechanism identification
34 """

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IPython Console Plots Variable Explorer Help Debugger History Files

C:\Users\Dell\Desktop\OPENSEES_FILES\+HEIGHT_ANALY...TEEL_PLATE_SECTION_DIFF_WIDTH_HEIGHT_FUN_EXTRA.py

P(t)_BEAM_ONE_ELEMENT_RELEASE_M3.py X I_STEEL_PLATE_SECT...EIGHT_FUN_EXTRA.py X

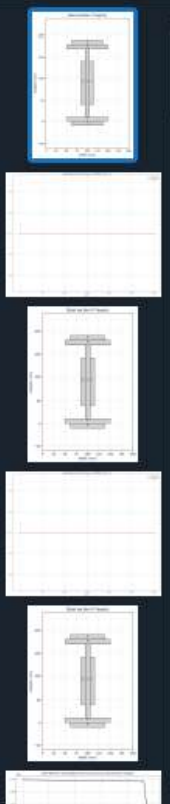
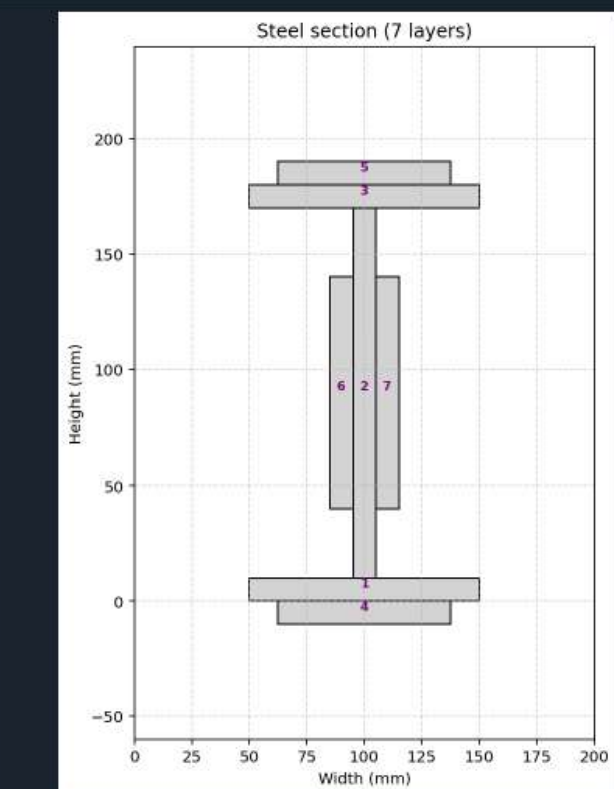
```

1 def I_STEEL_PLATE_SECTION_DIFF_WIDTH_HEIGHT_FUN_EXTRA(secTag, STEEL_TYPE, STEEL_DENSITY, plot=
2 """
3 Create a steel with different widths and heights, fiber section (OpenSees) and,
4 if PLOT=True, draw a simple 2-D sketch of the section.
5
6 Parameters
7 -----
8 secTag      : int - section identifier
9 STEEL_TYPE   : str - 'ELASTIC' or 'INELASTIC'
10 STEEL_DENSITY: float - pc (kg/m³) - will be converted to N·s²/mm³
11 PLOT        : bool - draw the section if True
12
13 THIS PYTHON SCRIPT IS WRITTEN BY SALAR DELAVAR GHASHGHAEE (QASHQAEI)
14 """
15 import numpy as np
16 import openseespy.opensees as ops
17 import matplotlib.pyplot as plt
18 import matplotlib.pyplot as plt, matplotlib.patches as patches
19
20 # Define materials for nonlinear elements
21 # Define parameters (units: mm, N)
22 # -----
23 # STEEL
24 fy = 240      # [N/mm²] Steel Yield Strength
25 Es = 2e5      # [N/mm²] Modulus of Elasticity
26 ey = fy/Es    # [mm/mm] Steel Yield Strain
27 fu = 1.1818*fy # [N/mm²] Steel Ultimate Strength
28 esu = 0.15    # [mm/mm] Steel Ultimate Strain
29 Esh = (fu - fy)/(esu - ey)
30 Bs = Esh / Es
31
32 steelTag = secTag + 100
33 if STEEL_TYPE == 'ELASTIC':

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STATIC ANALYSIS

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P(t)_BEAM_ONE_ELEMENT_RELEASE_M3.py X I_STEEL_PLATE_SECT...EIGHT_FUN_EXTRA.py X

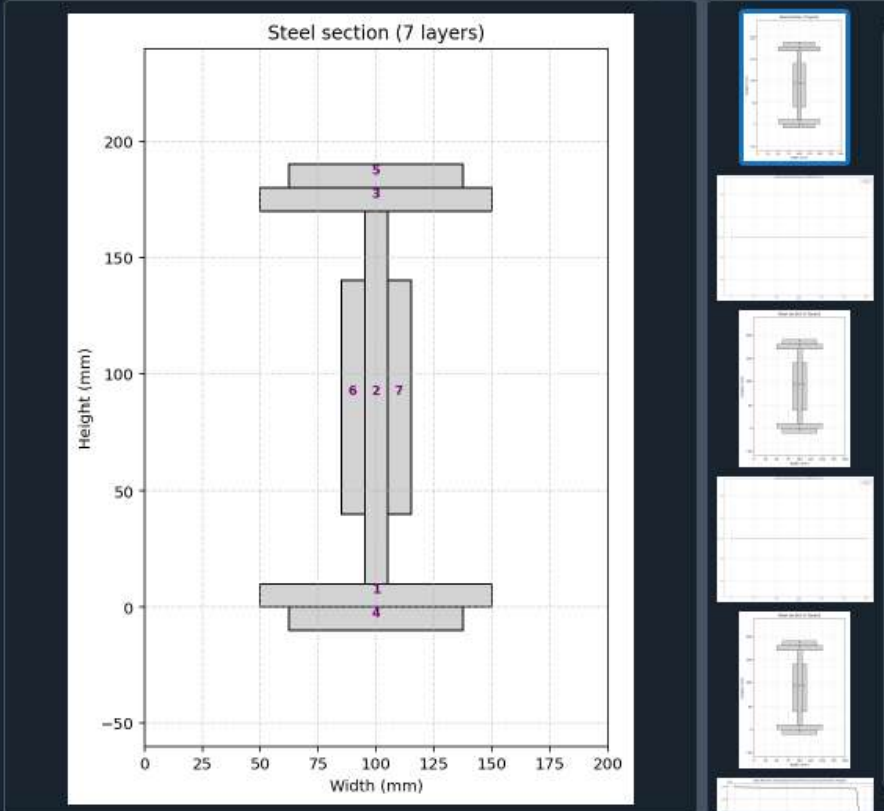
```

1034 # STATIC ANALYSIS
1035 #ELE_TYPE = 'elasticBeamColumn' # MATERIAL LINEARITY
1036 ELE_TYPE = 'nonlinearBeamColumn' # MATERIAL AND GEOMETRIC NONLINEARITIES
1037 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1038 ANAL_TYPE = 'STATIC'
1039
1040 DATA = BEAM_ONE_ELEMENT(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
1041 (reaction, disp_mid,
1042  ele_force, ele_stress, ele_strain, ele_di,
1043  node_displacementsX, node_displacementsY,
1044  dispX, dispY) = DATA
1045
1046 S01.PLOT_2D_FRAME(deformed scale=1.0)
1047
1048 #####
1049 # PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1050 #ELE_TYPE = 'elasticBeamColumn' # MATERIAL LINEARITY
1051 ELE_TYPE = 'nonlinearBeamColumn' # MATERIAL AND GEOMETRIC NONLINEARITIES
1052 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1053 ANAL_TYPE = 'PUSHOVER'
1054
1055 DATA = BEAM_ONE_ELEMENT(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
1056 (reaction_PUSH, disp_mid_PUSH,
1057  ele_force_PUSH, ele_stress_PUSH, ele_strain_PUSH, ele_di_PUSH,
1058  node_displacementsX_PUSH, node_displacementsY_PUSH,
1059  dispX_PUSH, dispY_PUSH,
1060  PERIOD_MIN_PUSH, PERIOD_MAX_PUSH,
1061  STRESSb_PUSH, STRAINb_PUSH, STRESSt_PUSH, STRAINT_PUSH, CURVATURE_PUSH) = DATA
1062
1063 XDATA = disp_mid_PUSH
1064 YDATA = reaction_PUSH
1065 XLABEL = 'Displacement [mm]'
1066 YLABEL = 'Base Reaction [N]'
1067 TITLE = 'Base Reaction and Displacement of Structure During Pushover Analysis'

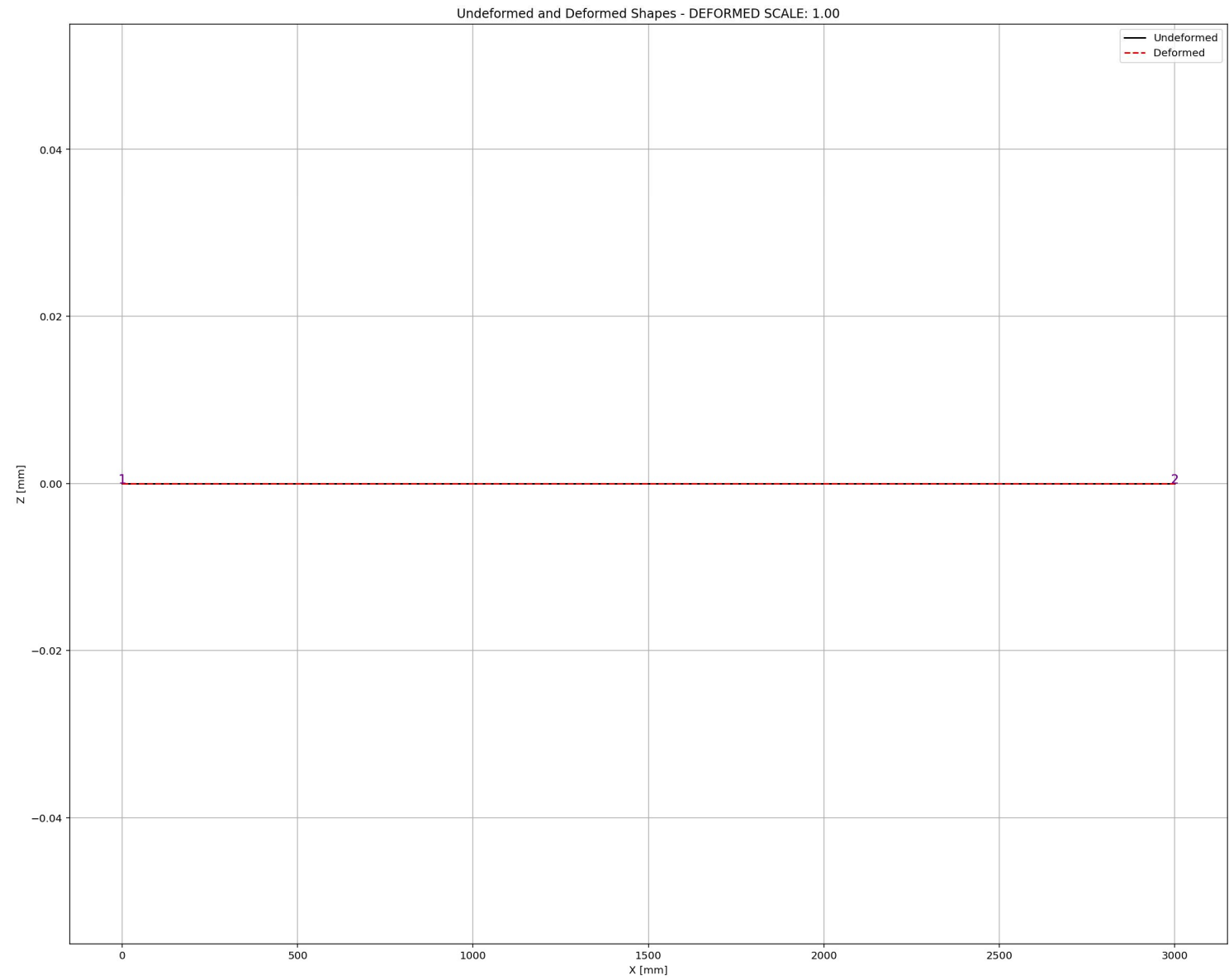
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PUSHOVER ANALYSIS

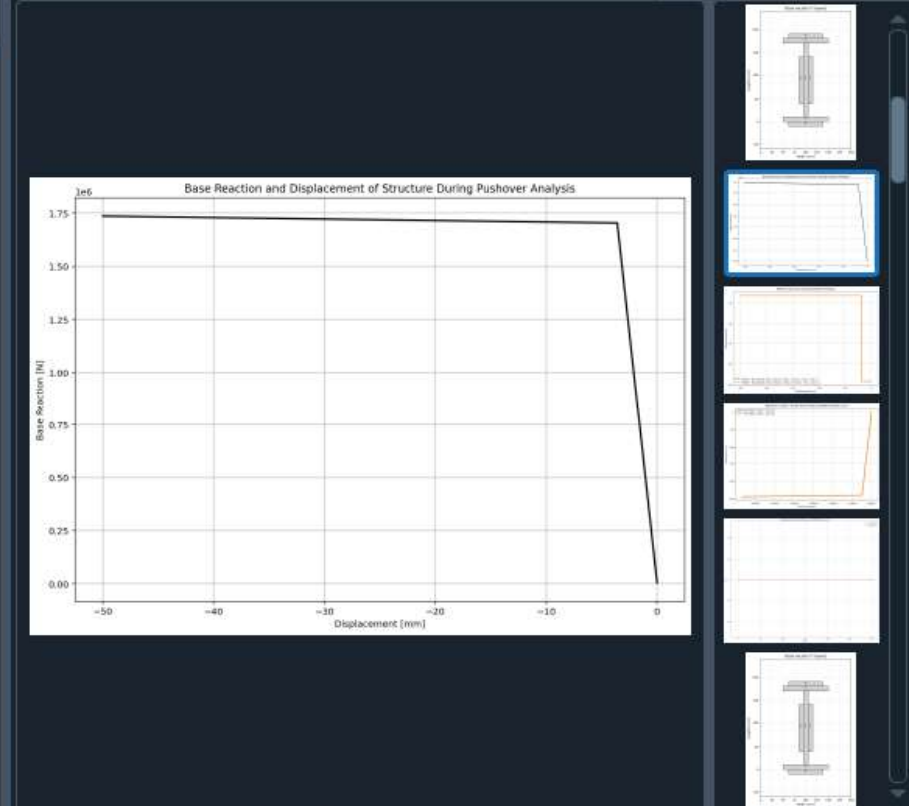
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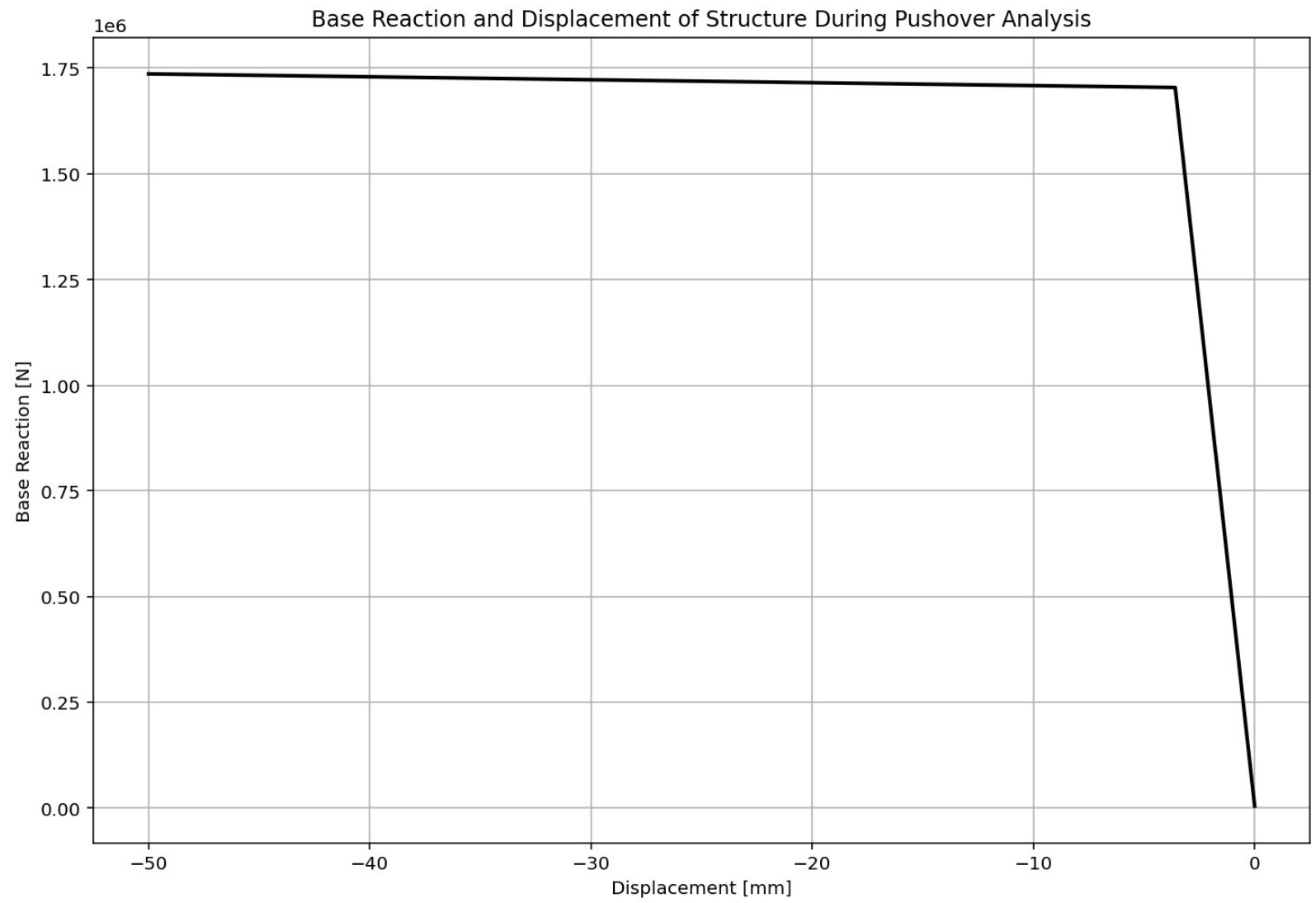
P(t)_BEAM_ONE_ELEMENT_RELEASE_M3.py X I_STEEL_PLATE_SECT...EIGHT_FUN_EXTRA.py X

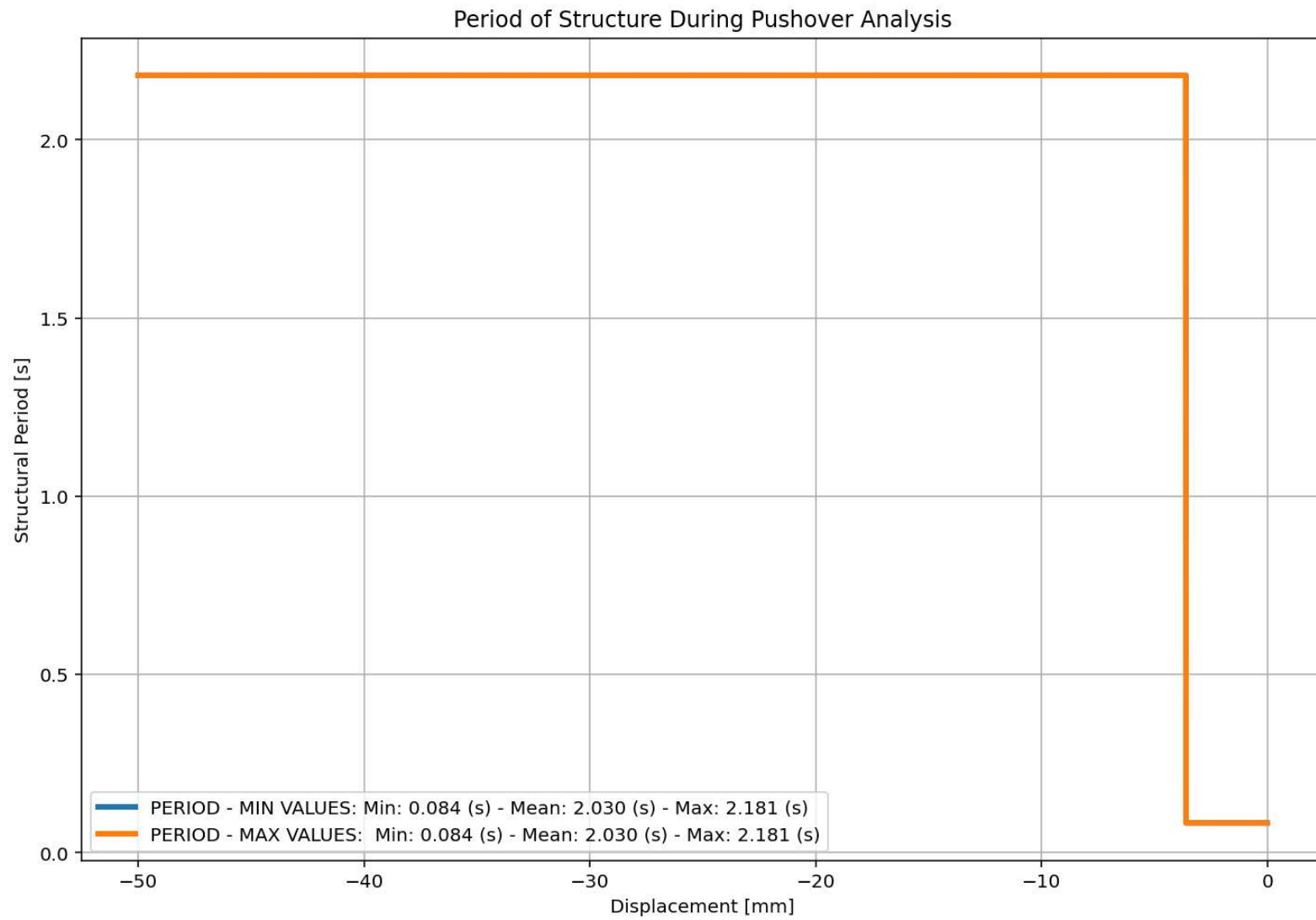
```

1048 # PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1049 #ELE_TYPE = 'elasticBeamColumn' # MATERIAL LINEARITY
1050 ELE_TYPE = 'nonlinearBeamColumn' # MATERIAL AND GEOMETRIC NONLINEARITIES
1051 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1052 ANAL_TYPE = 'PUSHOVER'
1053
1054 DATA = BEAM_ONE_ELEMENT(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
1055 (reaction_PUSH, disp_mid_PUSH,
1056  ele_force_PUSH, ele_stress_PUSH, ele_strain_PUSH, ele_di_PUSH,
1057  node_displacementsX_PUSH, node_displacementsY_PUSH,
1058  dispX_PUSH, dispY_PUSH,
1059  PERIOD_MIN_PUSH, PERIOD_MAX_PUSH,
1060  STRESSb_PUSH, STRAINb_PUSH, STRESSt_PUSH, STRAINT_PUSH, CURVATURE_PUSH) = DATA
1061
1062
1063 XDATA = disp_mid_PUSH
1064 YDATA = reaction_PUSH
1065 XLABEL = 'Displacement [mm]'
1066 YLABEL = 'Base Reaction [N]'
1067 TITLE = 'Base Reaction and Displacement of Structure During Pushover Analysis'
1068 COLOR = 'black'
1069 SEMILOGY = False
1070 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1071
1072 DATA = S07.BILNEAR_CURVE(np.abs(disp_mid_PUSH), np.abs(reaction_PUSH), SLOPE_NODE=10)
1073 (X_PUSH, Y_PUSH, Elastic_ST, Plastic_ST, Tangent_ST, Ductility_Rito, Over_Strength_Factor) =
1074
1075 # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1076 plt.figure(0, figsize=(12, 8))
1077 plt.plot(disp_mid_PUSH, PERIOD_MIN_PUSH, linewidth=3)
1078 plt.plot(disp_mid_PUSH, PERIOD_MAX_PUSH, linewidth=3)
1079 plt.title('Period of Structure During Pushover Analysis')
1080 plt.ylabel('Structural Period [s]')
1081 plt.xlabel('Displacement [mm]')
    
```

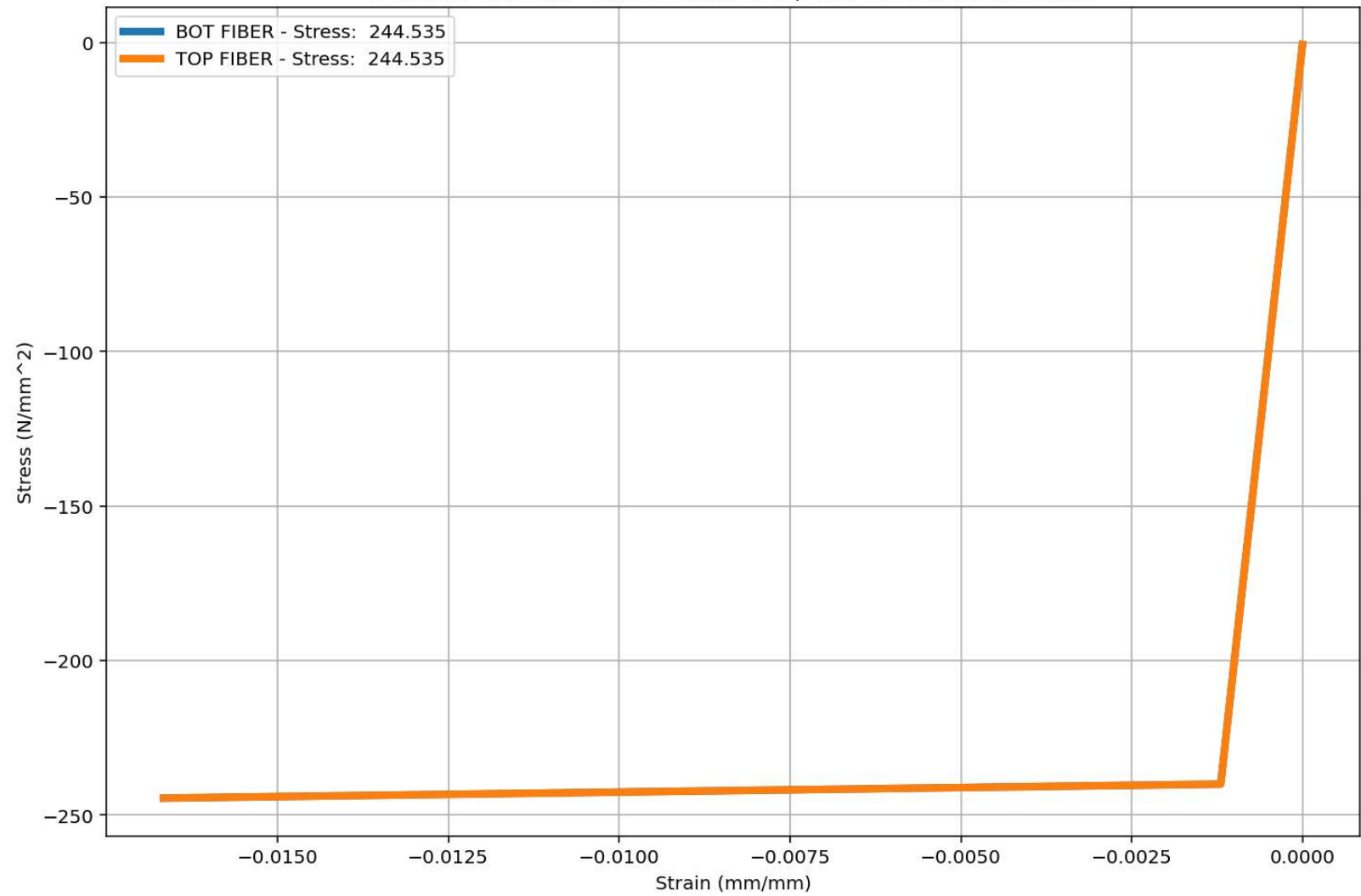
29 %

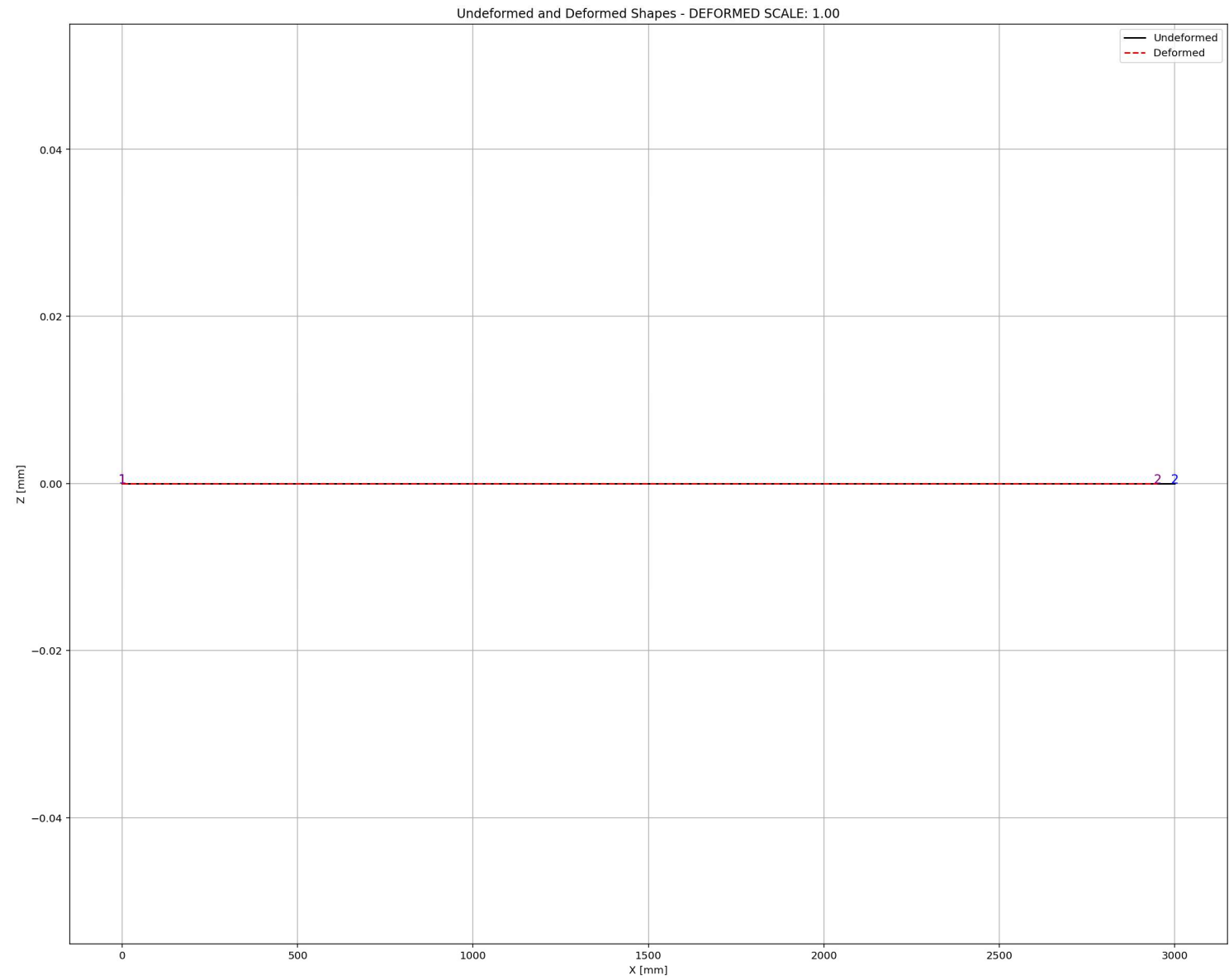






Behavior of beam - Stress-strain of Top and Bottom Fibers Curve



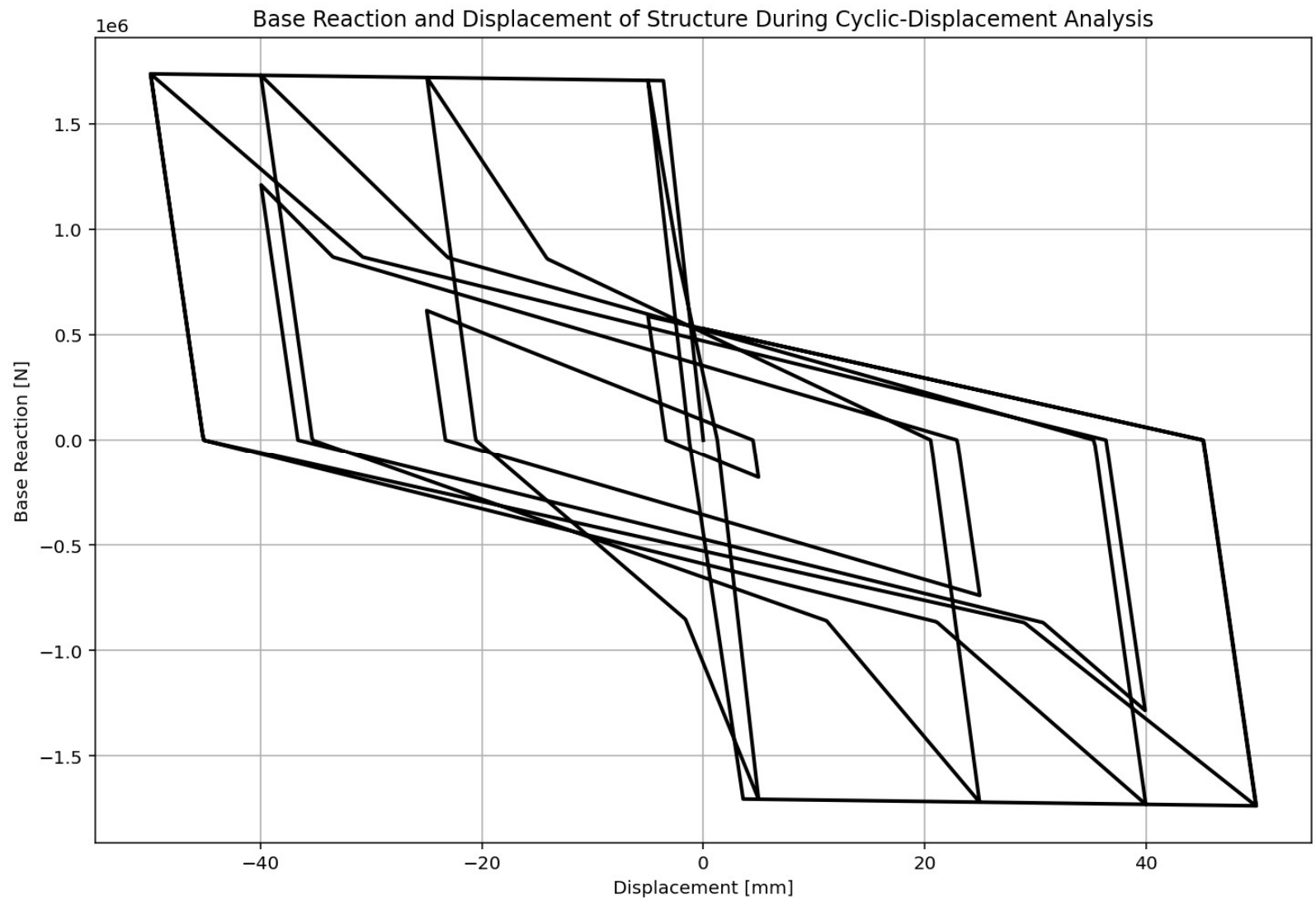


CYCLIC DISPLACEMENT ANALYSIS

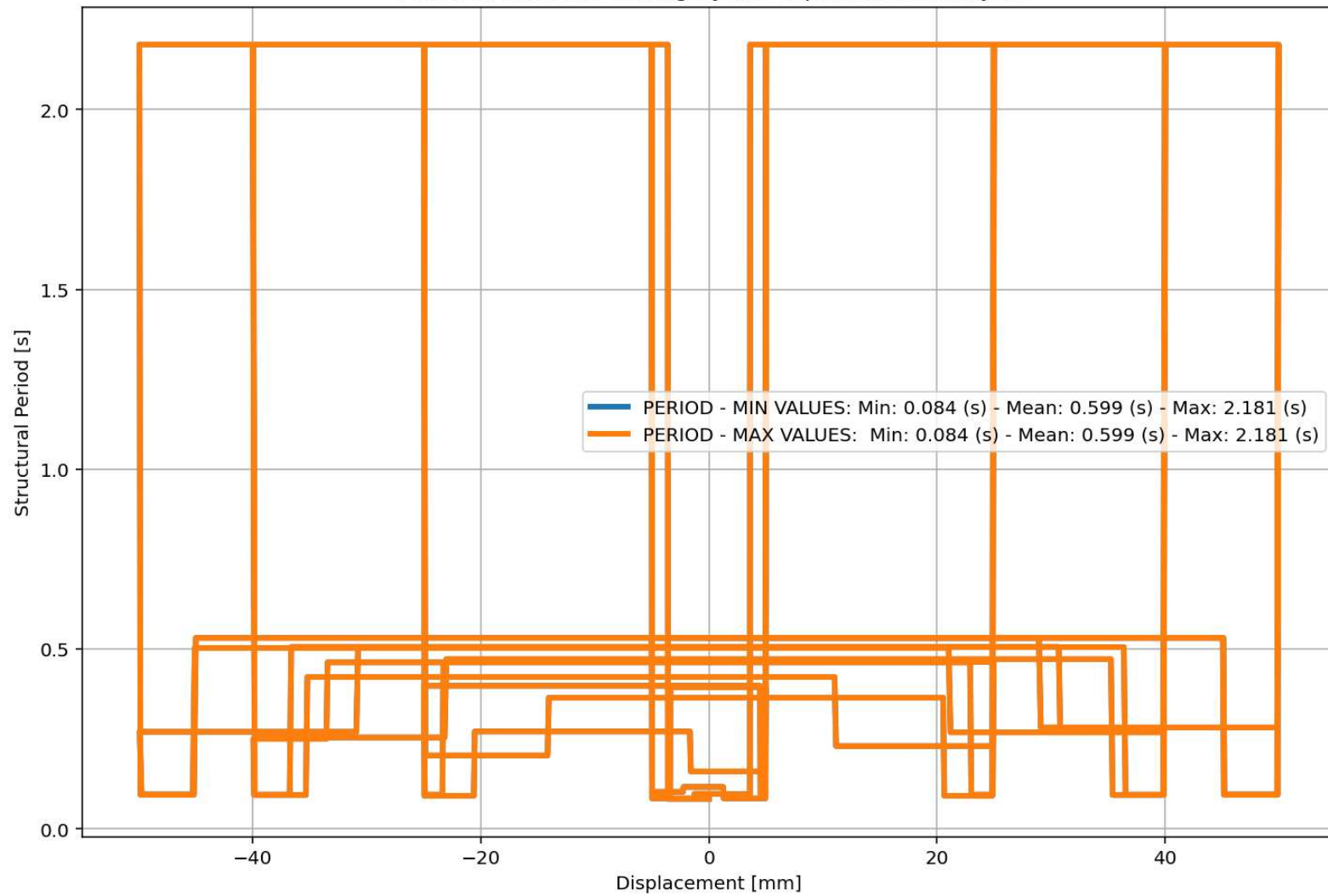
The main plot, titled "Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis", shows the relationship between base reaction and displacement. The y-axis is labeled "Base Reaction (kN)" and ranges from -1.5 to 1.5, with a multiplier of $\times 10^6$ at the top. The x-axis is labeled "Displacement (mm)" and ranges from -40 to 40. The plot displays multiple hysteresis loops, indicating the structure's behavior under cyclic loading. The loops are generally rectangular, suggesting a ductile response, with some pinching observed at lower displacement levels.

On the right side, there is a vertical strip of five smaller images:

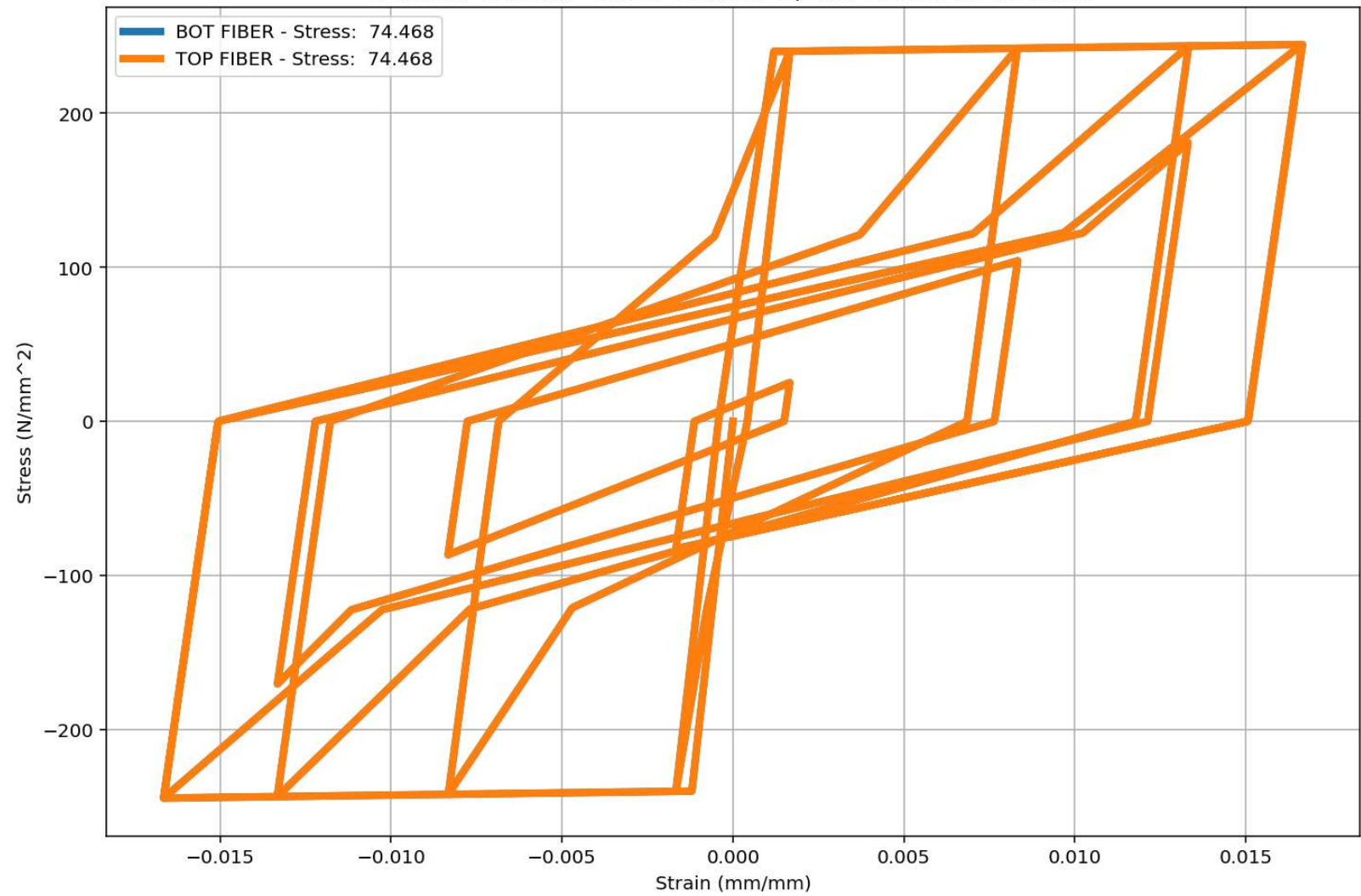
- The top image shows a 3D wireframe model of a structural frame.
- The second image shows a 2D cross-section of a structural member, possibly a beam or column, with internal reinforcement details.
- The third image shows a 3D model of a structural frame, similar to the top image but with different shading or highlighting.
- The fourth image shows a 2D cross-section of a structural member, similar to the second image, with a red shaded area indicating a specific region of interest or damage.
- The bottom image shows a 2D cross-section of a structural member, similar to the others, with a red shaded area.



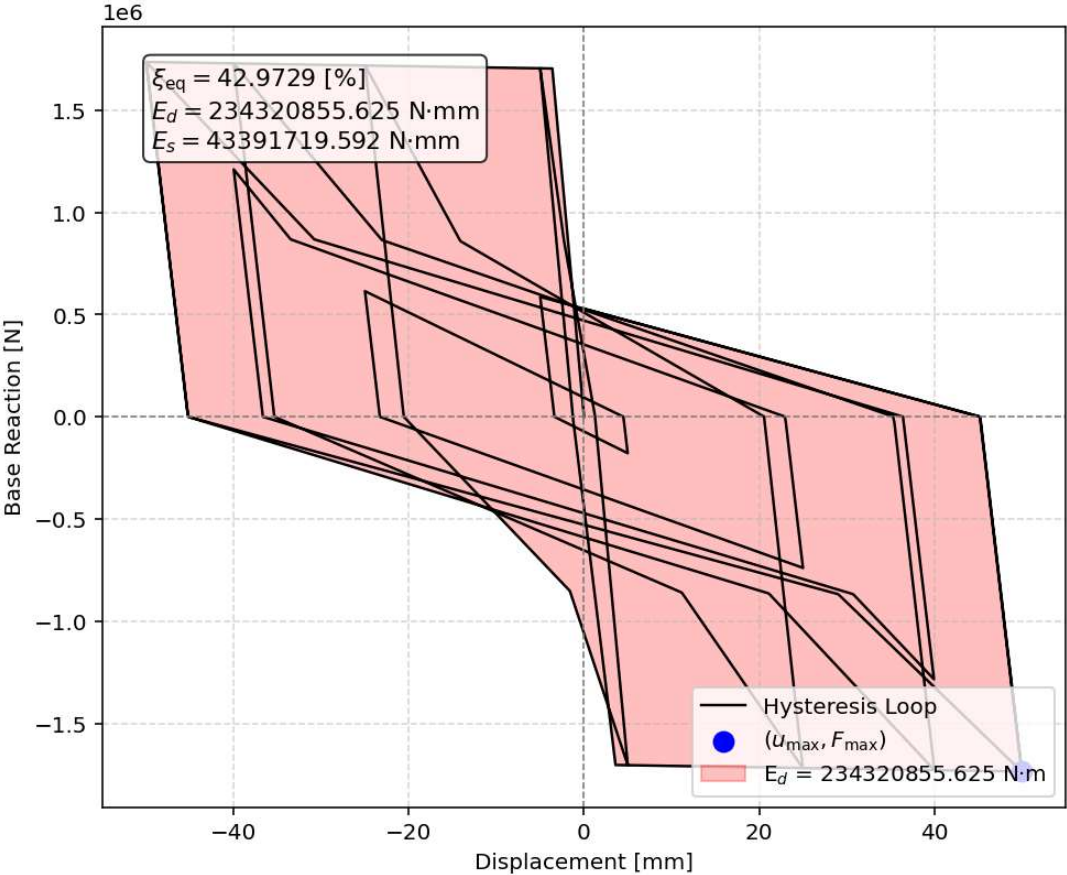
Period of Structure During Cyclic Displacement Analysis

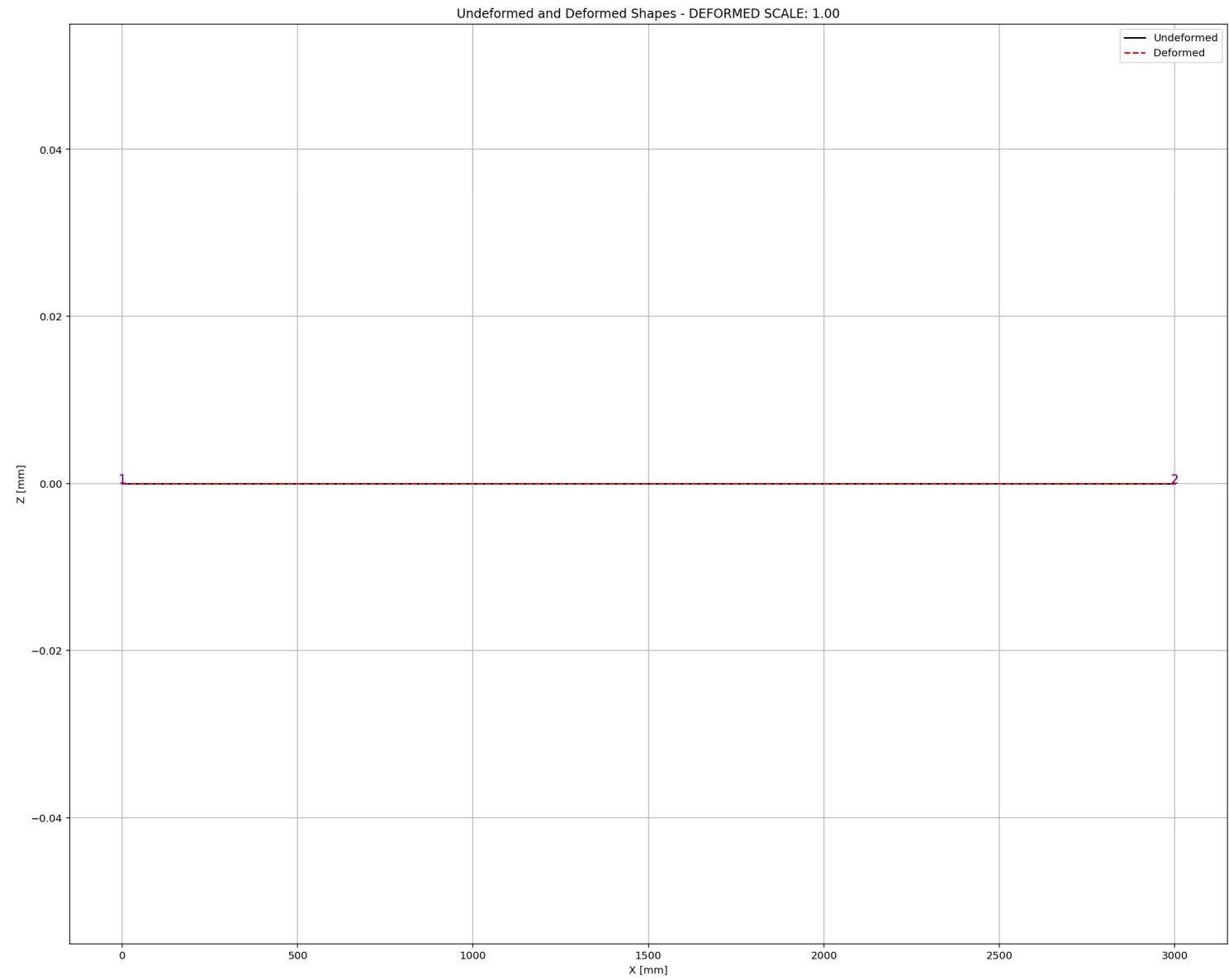


Behavior of beam - Stress-strain of Top and Bottom Fibers Curve



Equivalent viscous damping ratio - Cyclic Displacement Hysteresis





STATIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

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p(t)_BEAM_ONE_ELEMENT_RELEASE_M3.py x I_STEEL_PLATE_SECT...EIGHT_FUN_EXTRA.py x

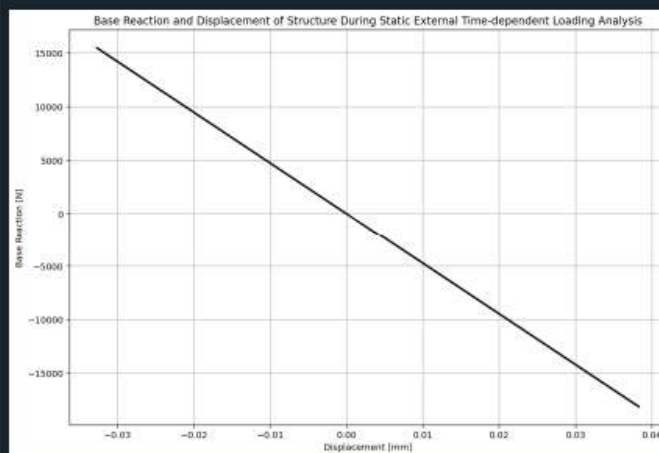
```

1160 # EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1161 #ELE_TYPE = 'elasticBeamColumn' # MATERIAL LINEARITY
1162 ELE_TYPE = 'nonlinearBeamColumn' # MATERIAL AND GEOMETRIC NONLINEARITIES
1163 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1164 ANAL_TYPE = 'STATIC_EXTERNAL_TIME-DEPENDENT_LOADING'
1165
1166 DATA = BEAM_ONE_ELEMENT(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
1167
1168 (reaction_ETDLS, disp_mid_ETDLS,
1169  ele_force_ETDLS, ele_stress_ETDLS, ele_strain_ETDLS, ele_di_ETDLS,
1170  node_displacementsX_ETDLS, node_displacementsY_ETDLS,
1171  dispX_ETDLS, dispY_ETDLS,
1172  PERIOD_MIN_ETDLS, PERIOD_MAX_ETDLS,
1173  STRESSb_ETDLS, STRAINb_ETDLS, STRESSt_ETDLS, STRAINt_ETDLS, CURVATURE_ETDLS) = DATA
1174
1175
1176 XDATA = disp_mid_ETDLS
1177 YDATA = reaction_ETDLS
1178 XLABEL = 'Displacement [mm]'
1179 YLABEL = 'Base Reaction [N]'
1180 TITLE = 'Base Reaction and Displacement of Structure During Static External Time-dependent Lo
1181 COLOR = 'black'
1182 SEMIOLOGY = False
1183 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMIOLOGY)
1184
1185 # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1186 plt.figure(0, figsize=(12, 8))
1187 plt.plot(disp_mid_ETDLS, PERIOD_MIN_ETDLS, linewidth=3)
1188 plt.plot(disp_mid_ETDLS, PERIOD_MAX_ETDLS, linewidth=3)
1189 plt.title('Period of Structure During Static External Time-dependent Loading Analysis')
1190 plt.ylabel('Structural Period [s]')
1191 plt.xlabel('Displacement [mm]')
1192 #plt.semilogy()
1193 plt.grid()

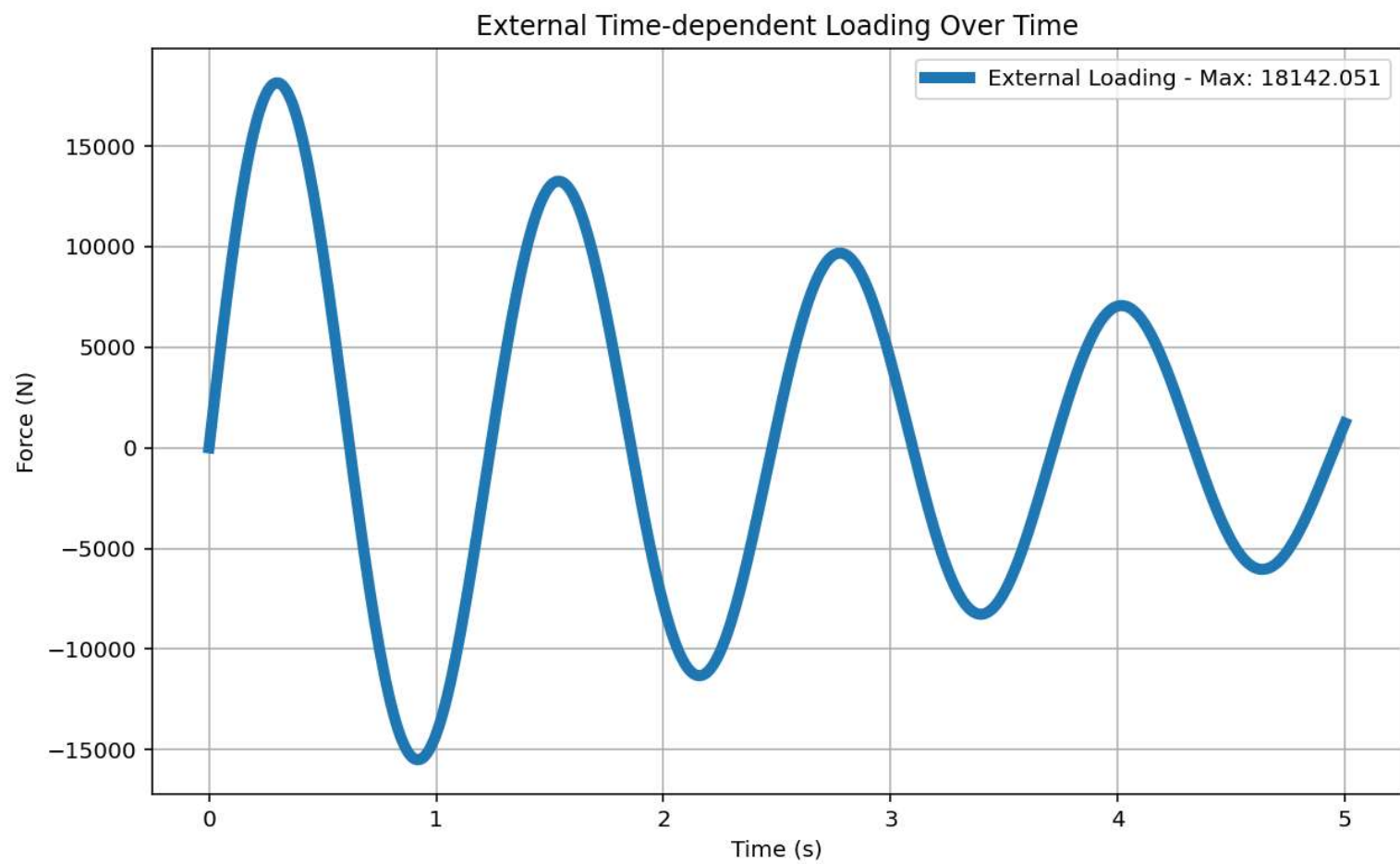
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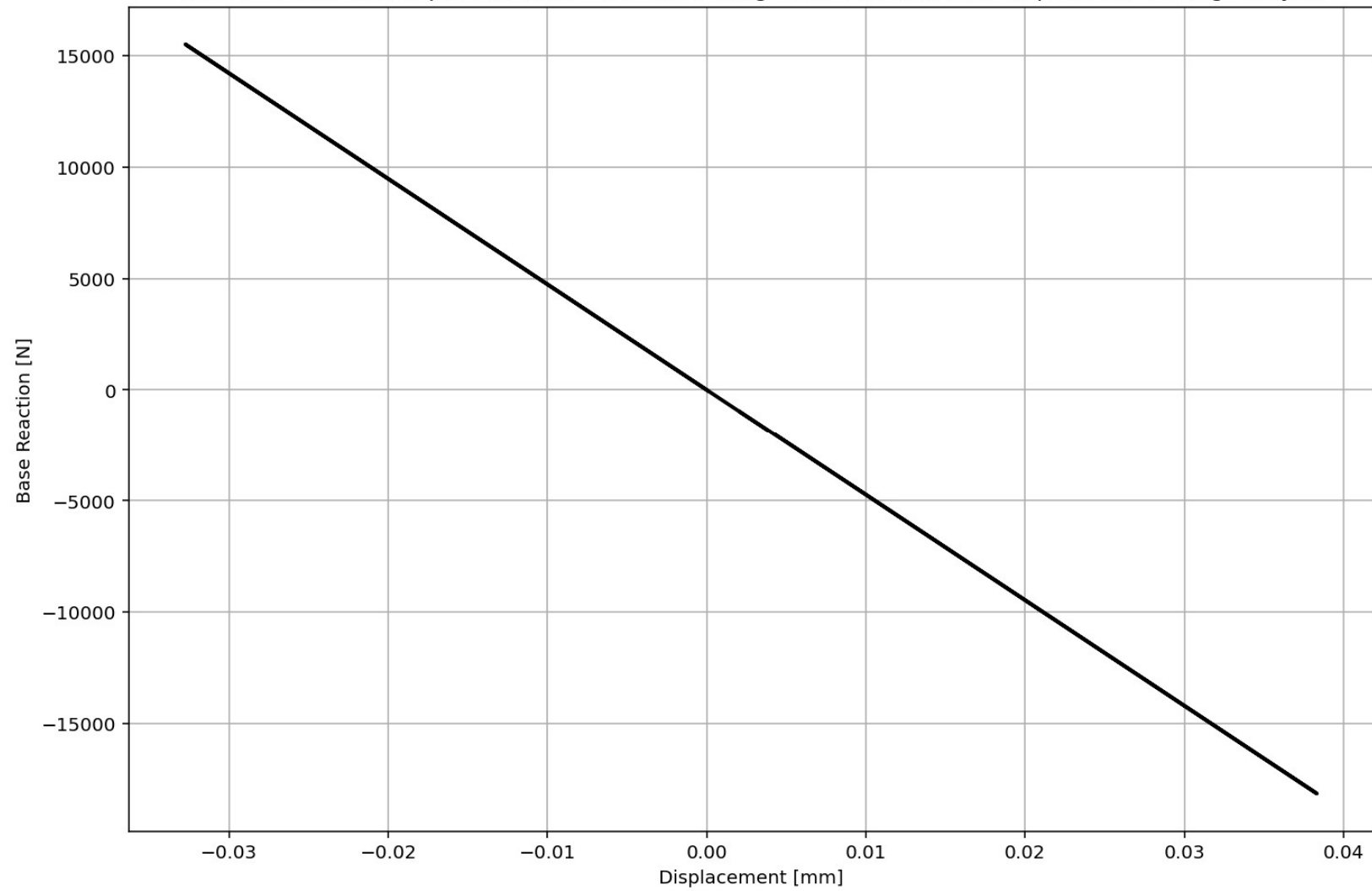
29 %



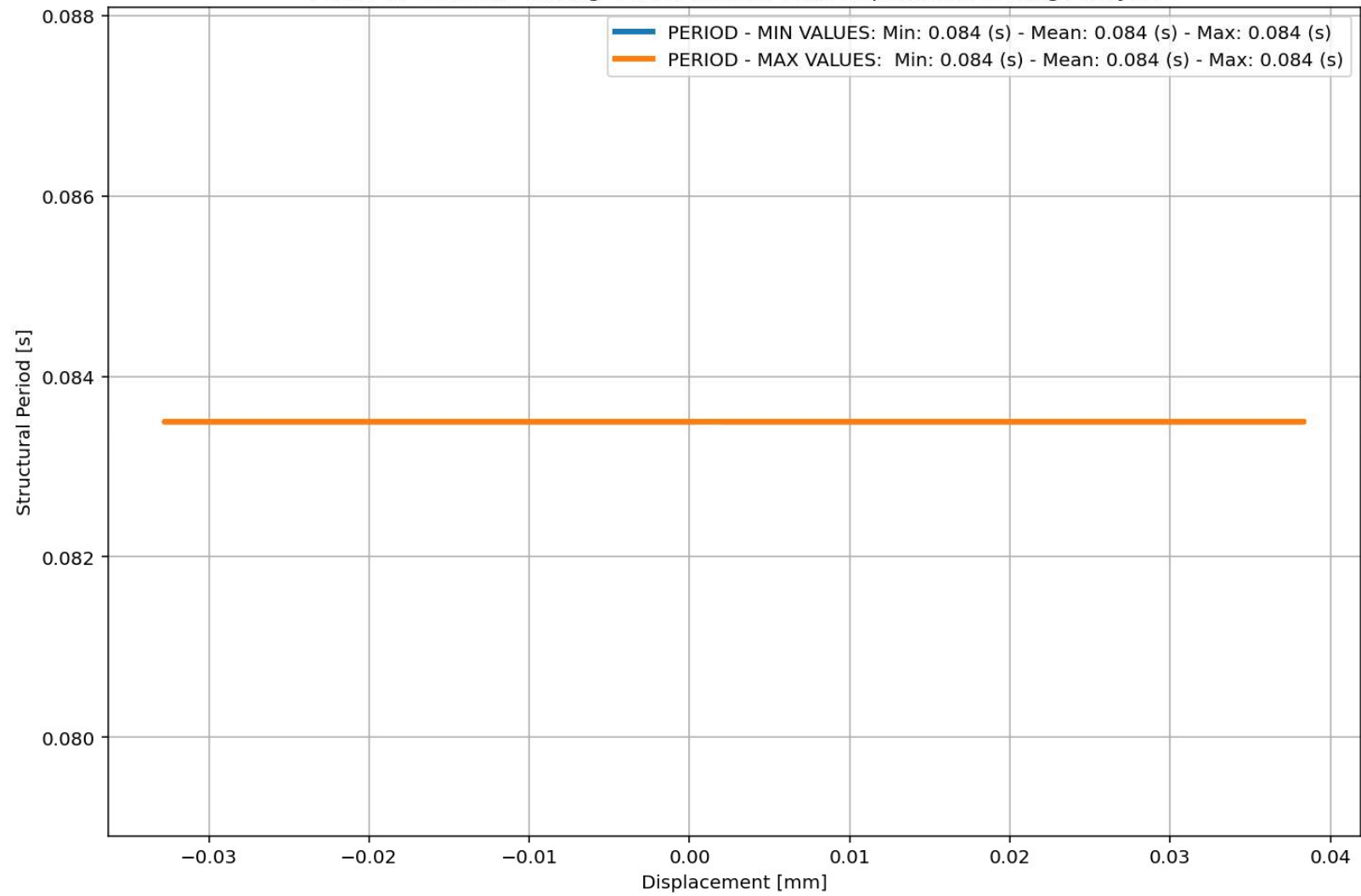
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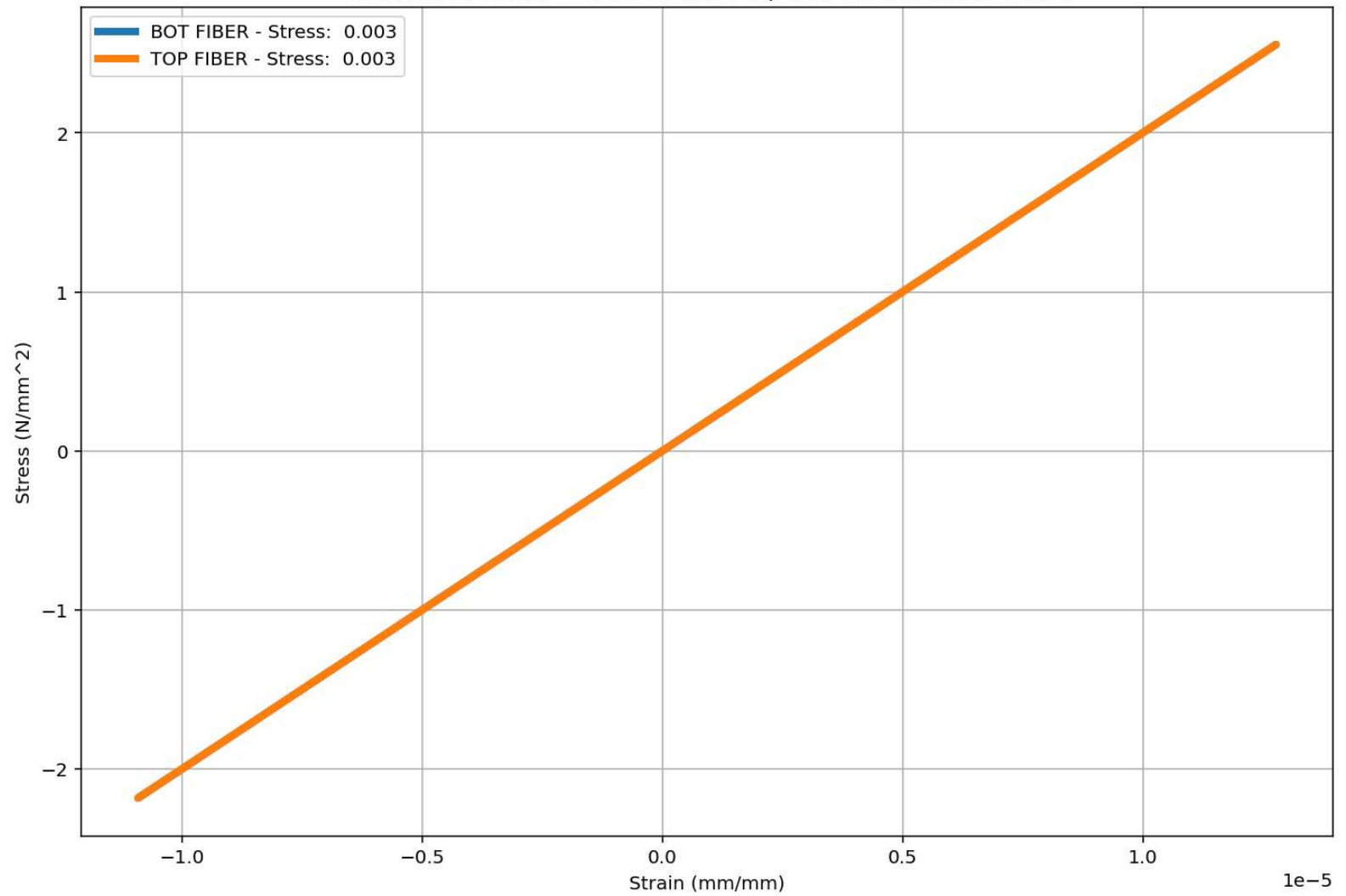
Base Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis

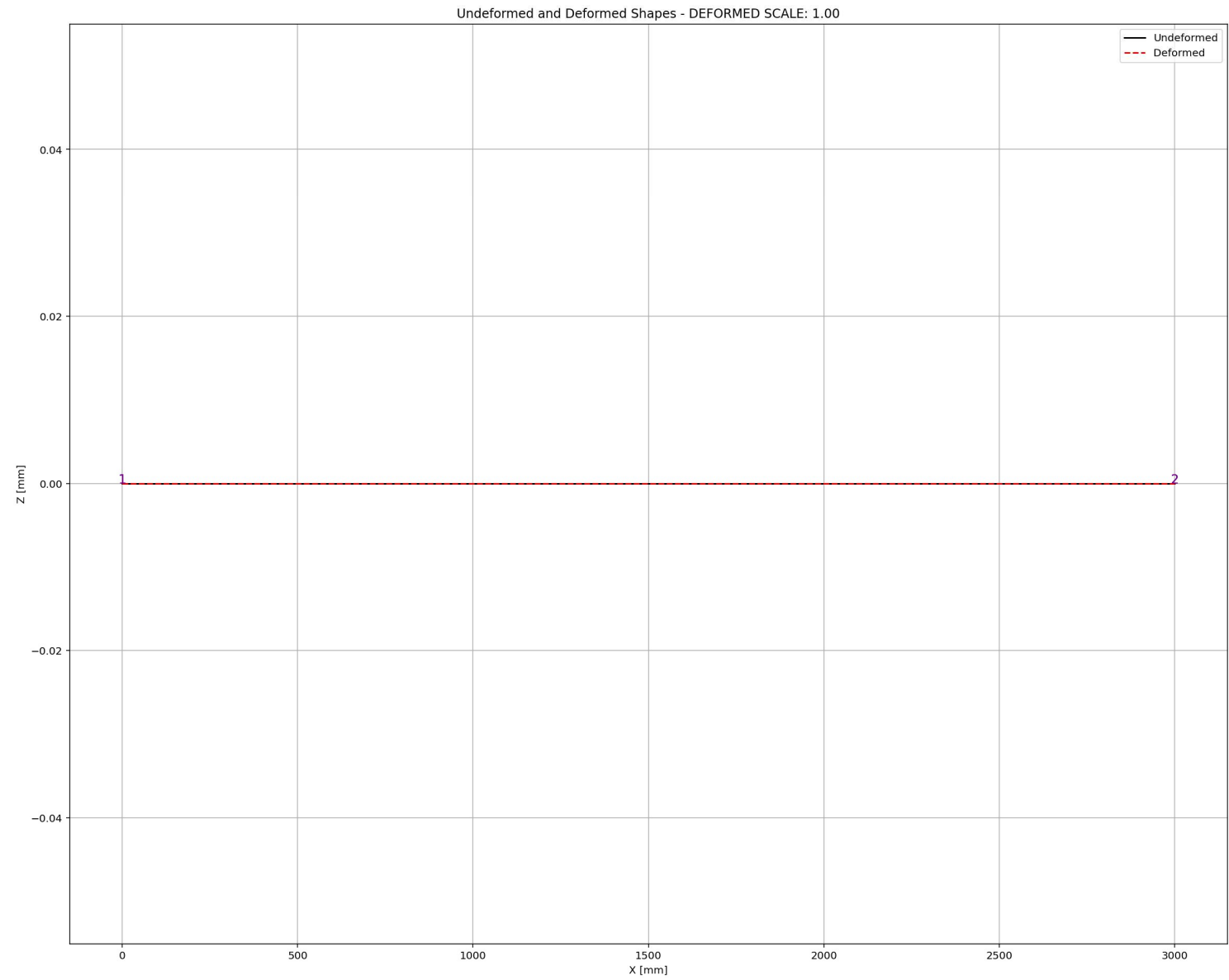


Period of Structure During Static External Time-dependent Loading Analysis



Behavior of beam - Stress-strain of Top and Bottom Fibers Curve





DYNAMIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

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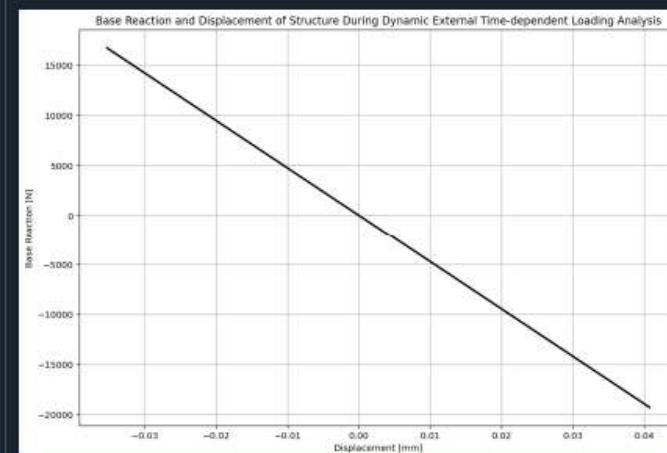
```

1212 # EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
1213 #ELE_TYPE = 'elasticBeamColumn' # MATERIAL LINEARITY
1214 ELE_TYPE = 'nonlinearBeamColumn' # MATERIAL AND GEOMETRIC NONLINEARITIES
1215 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1216 ANAL_TYPE = 'DYNAMIC_EXTERNAL_TIME-DEPENDENT_LOADING'
1217
1218 DATA = BEAM_ONE_ELEMENT(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
1219
1220 (time_ETDLD, reaction_ETDLD, disp_mid_ETDLD,
1221 ele_force_ETDLD, ele_stress_ETDLD, ele_strain_ETDLD, ele_di_ETDLD,
1222 node_displacementsX_ETDLD, node_displacementsY_ETDLD,
1223 dispX_ETDLD, dispY_ETDLD,
1224 veloX_ETDLD, veloY_ETDLD,
1225 accX_ETDLD, accY_ETDLD,
1226 stiffness_ETDLD, PERIOD_ETDLD, damping_ratio_ETDLD,
1227 PERIOD_MIN_ETDLD, PERIOD_MAX_ETDLD,
1228 STRESSb_ETDLD, STRAINb_ETDLD, STRESSt_ETDLD, STRAINT_ETDLD, CURVATURE_ETDLD) = DATA
1229
1230
1231 XDATA = disp_mid_ETDLD
1232 YDATA = reaction_ETDLD
1233 XLABEL = 'Displacement [mm]'
1234 YLABEL = 'Base Reaction [N]'
1235 TITLE = 'Base Reaction and Displacement of Structure During Dynamic External Time-dependent L
1236 COLOR = 'black'
1237 SEMILOGY = False
1238 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1239
1240 PLOT_TIME_HISTORY(time_ETDLD, reaction_ETDLD, disp_mid_ETDLD,
1241 dispX_ETDLD, dispY_ETDLD,
1242 veloX_ETDLD, veloY_ETDLD,
1243 accX_ETDLD, accY_ETDLD)
1244
1245 # PLOT ELEMENTS AXIAL FORCE

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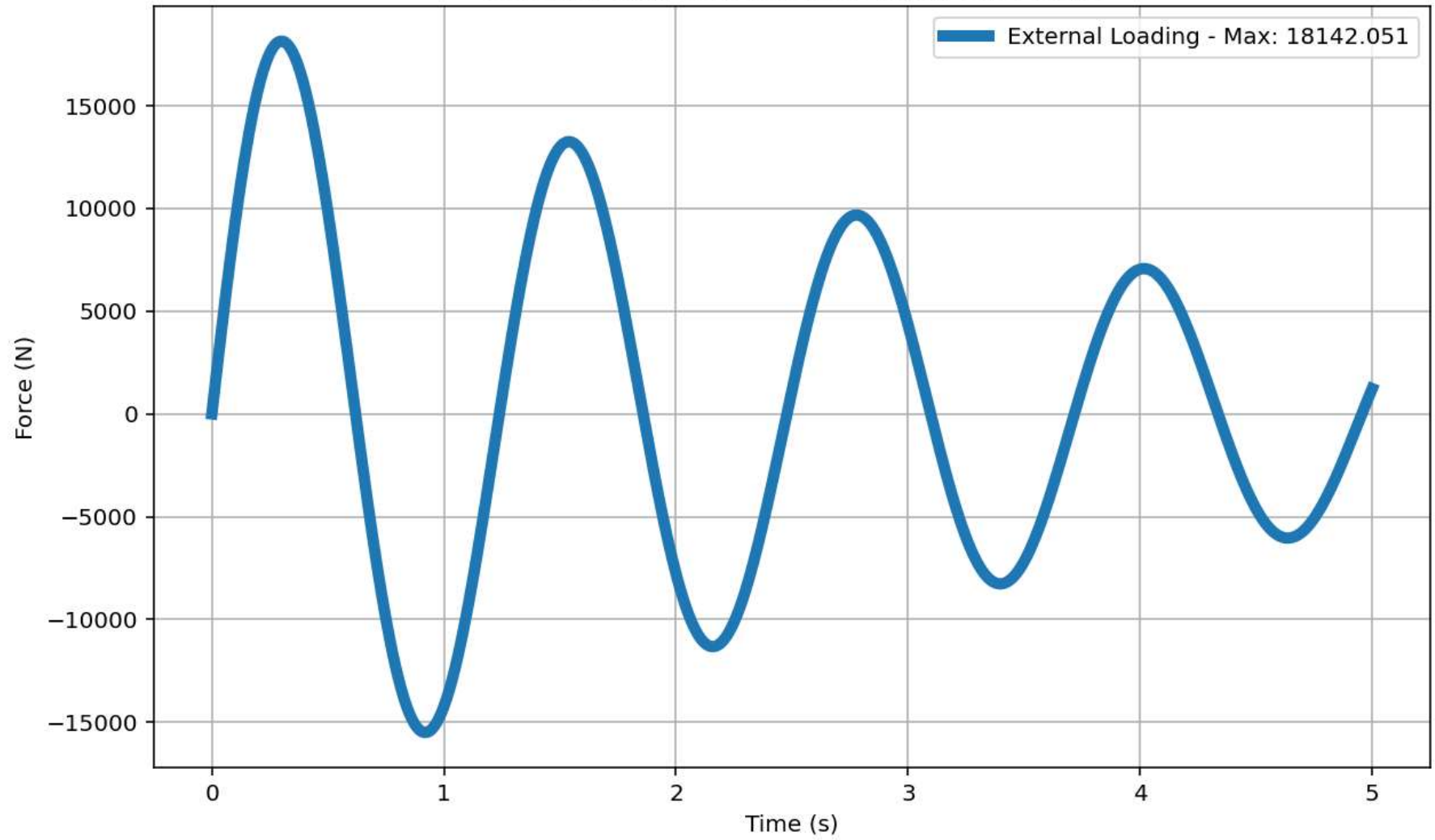
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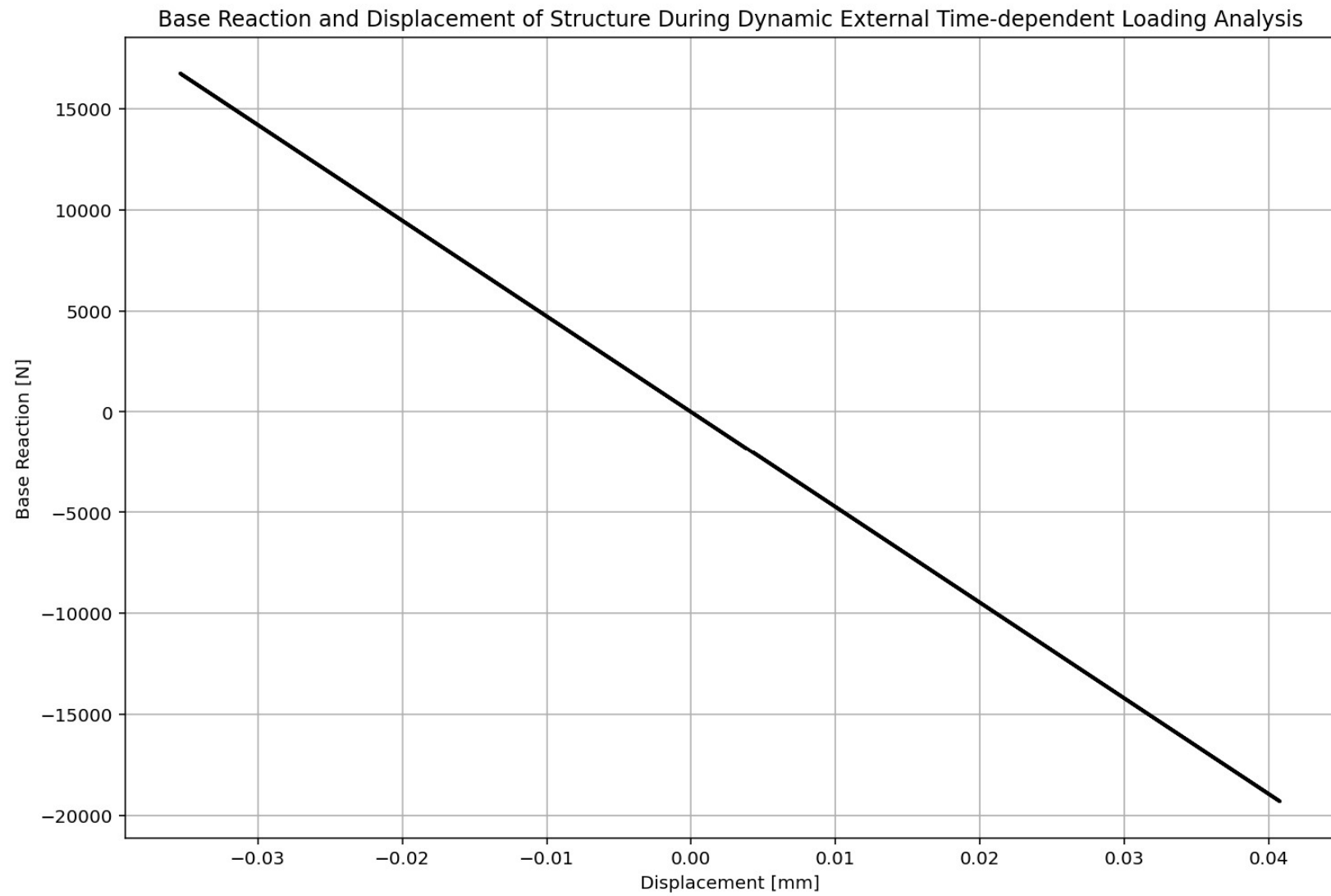
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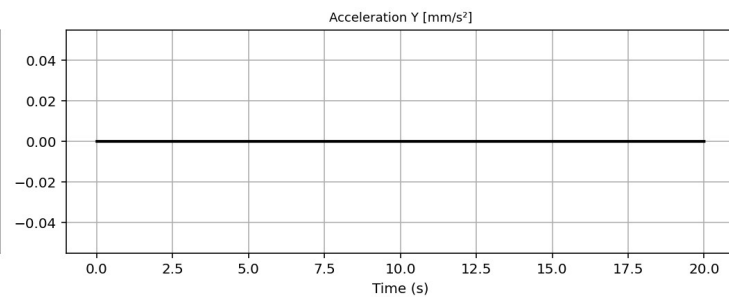
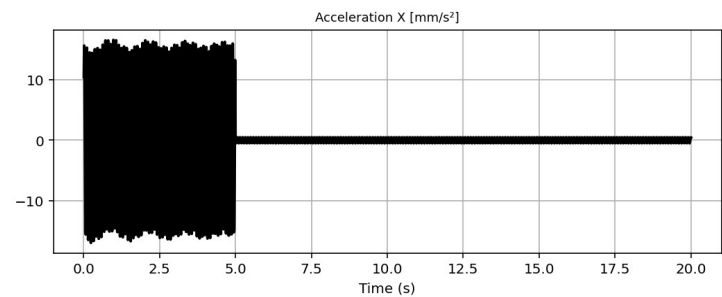
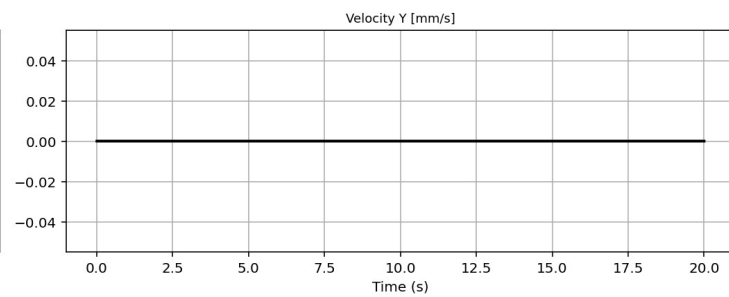
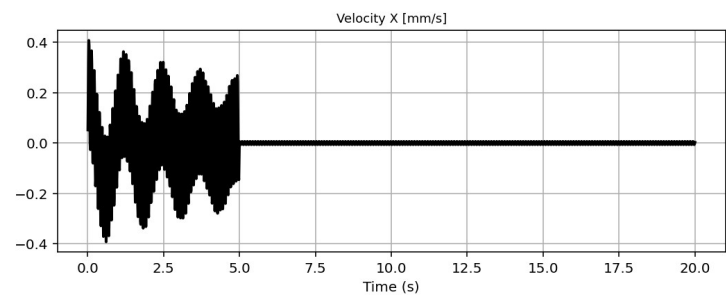
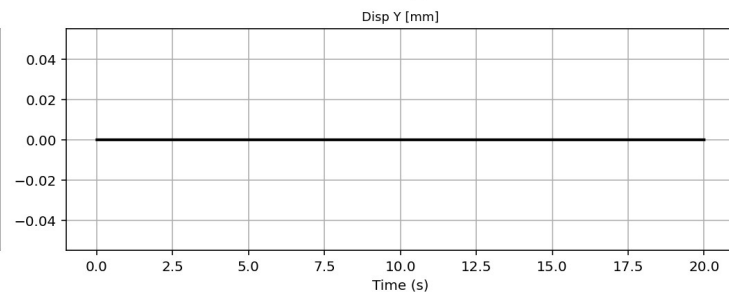
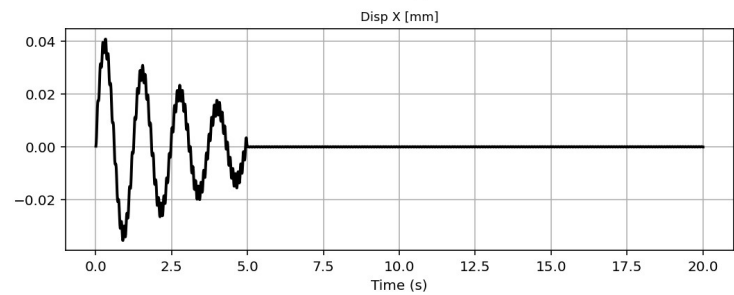
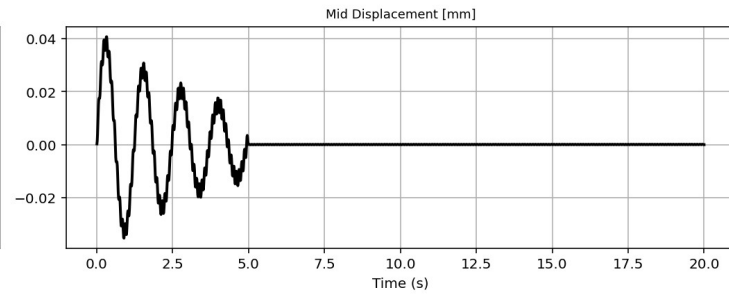
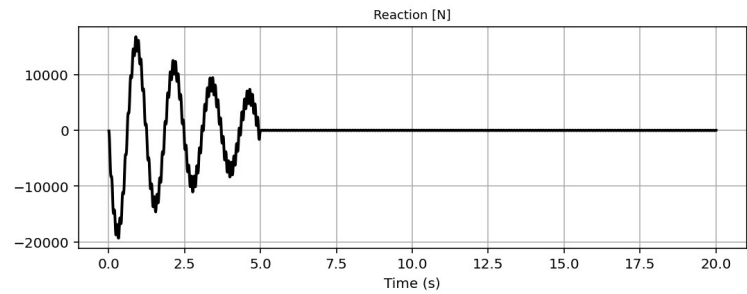


IPython Console Plots Variable Explorer Help Debugger History Files

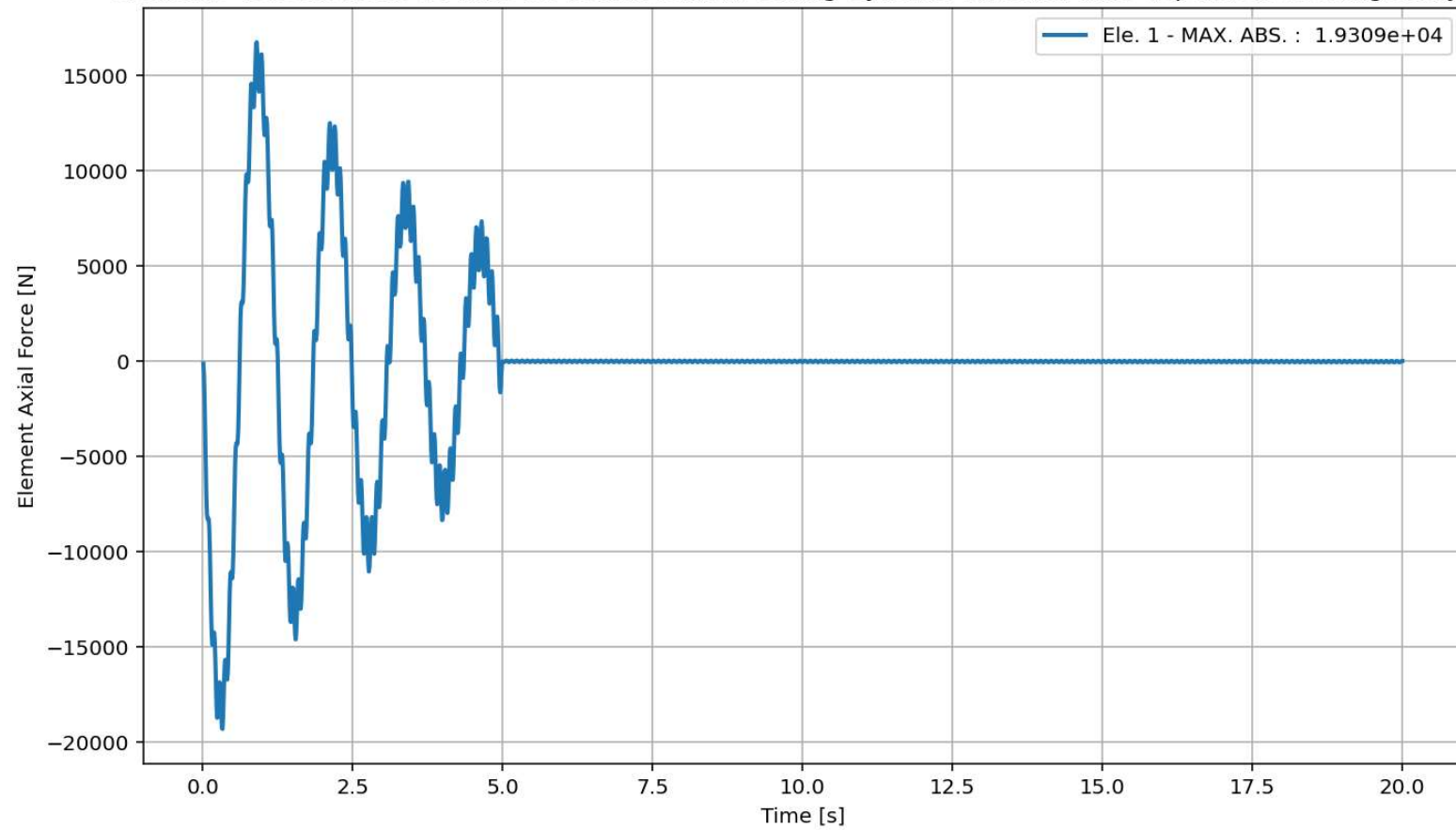
External Time-dependent Loading Over Time



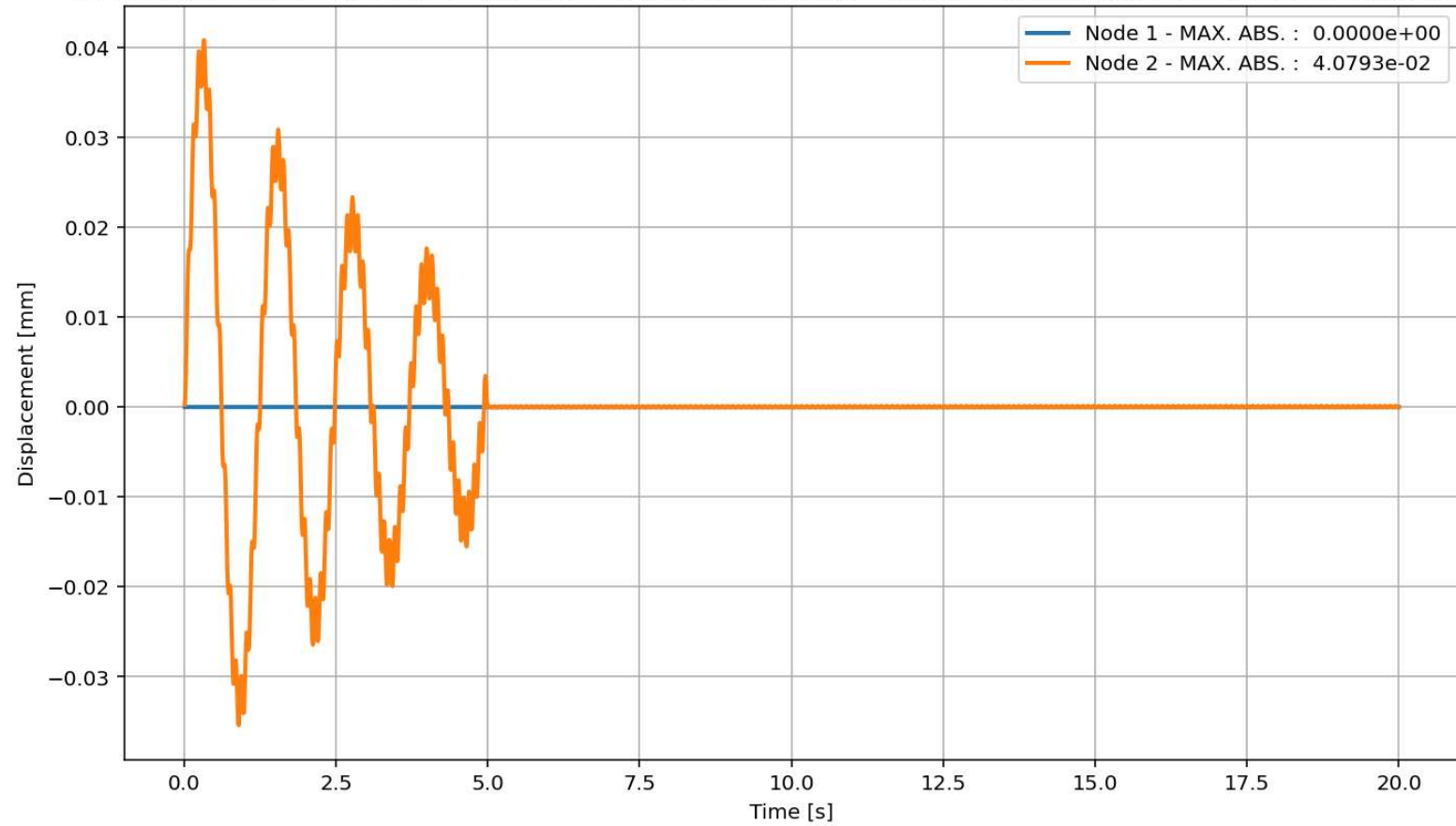




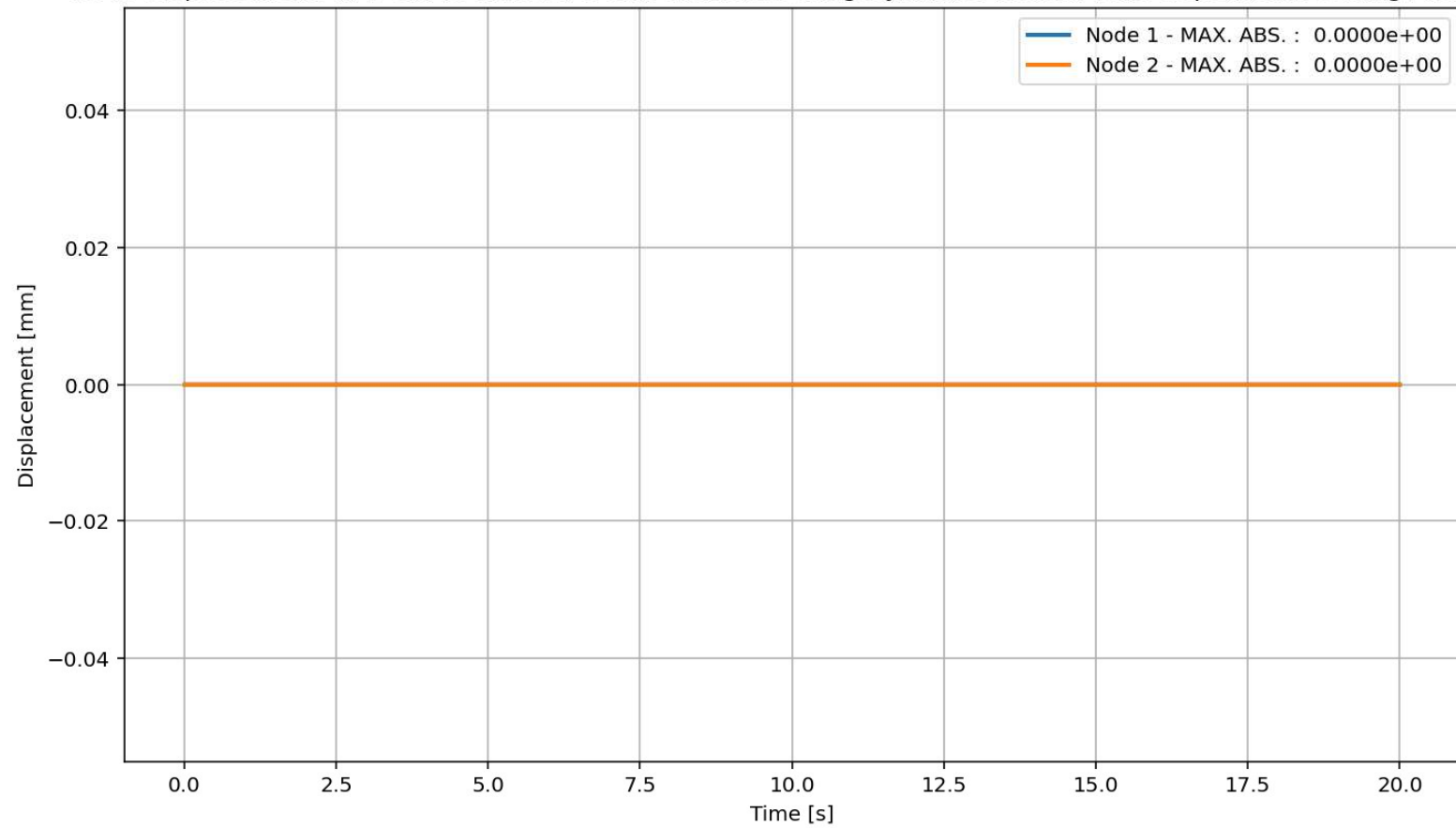
elements force in X Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis

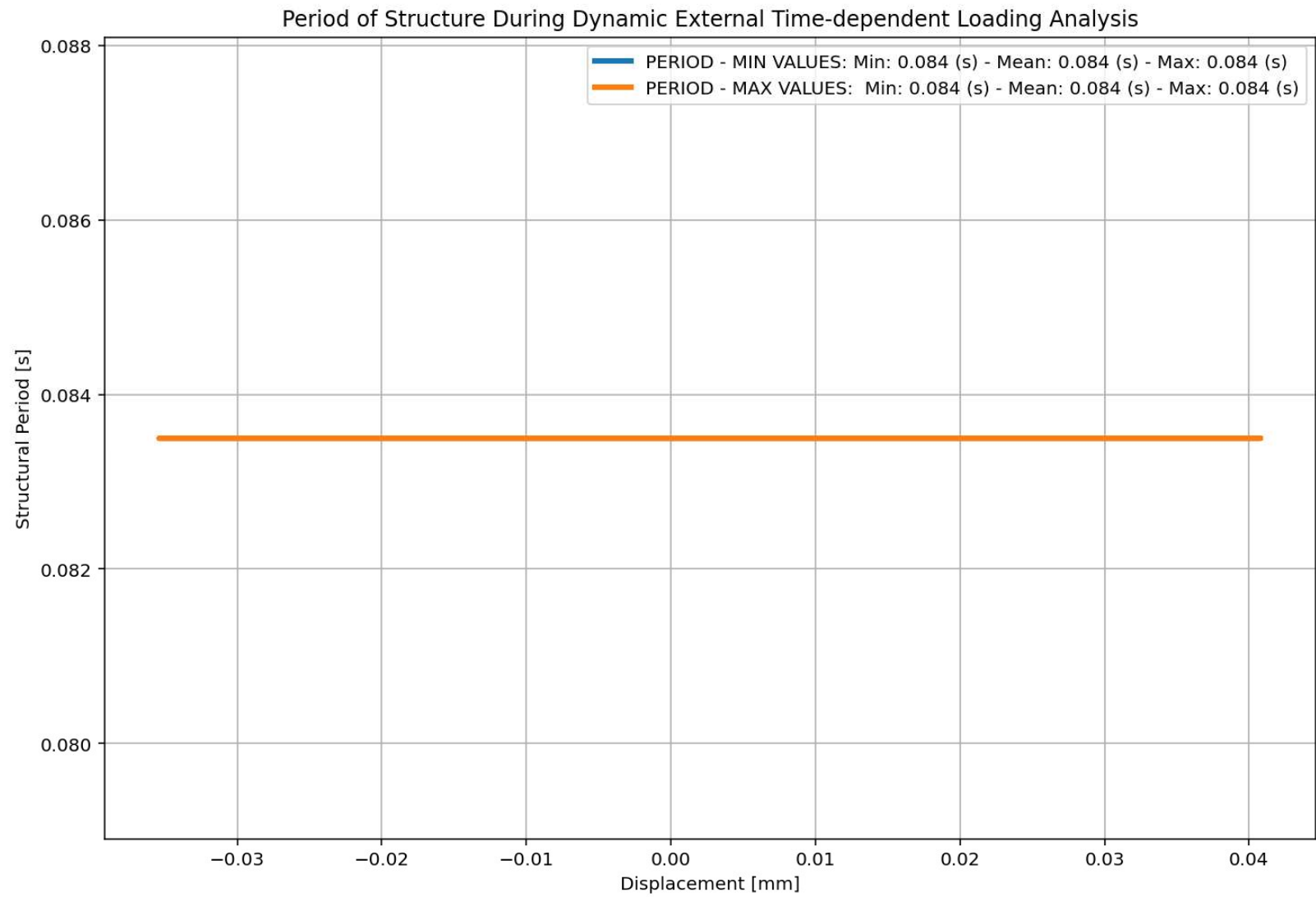


Node Displacements in X Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis

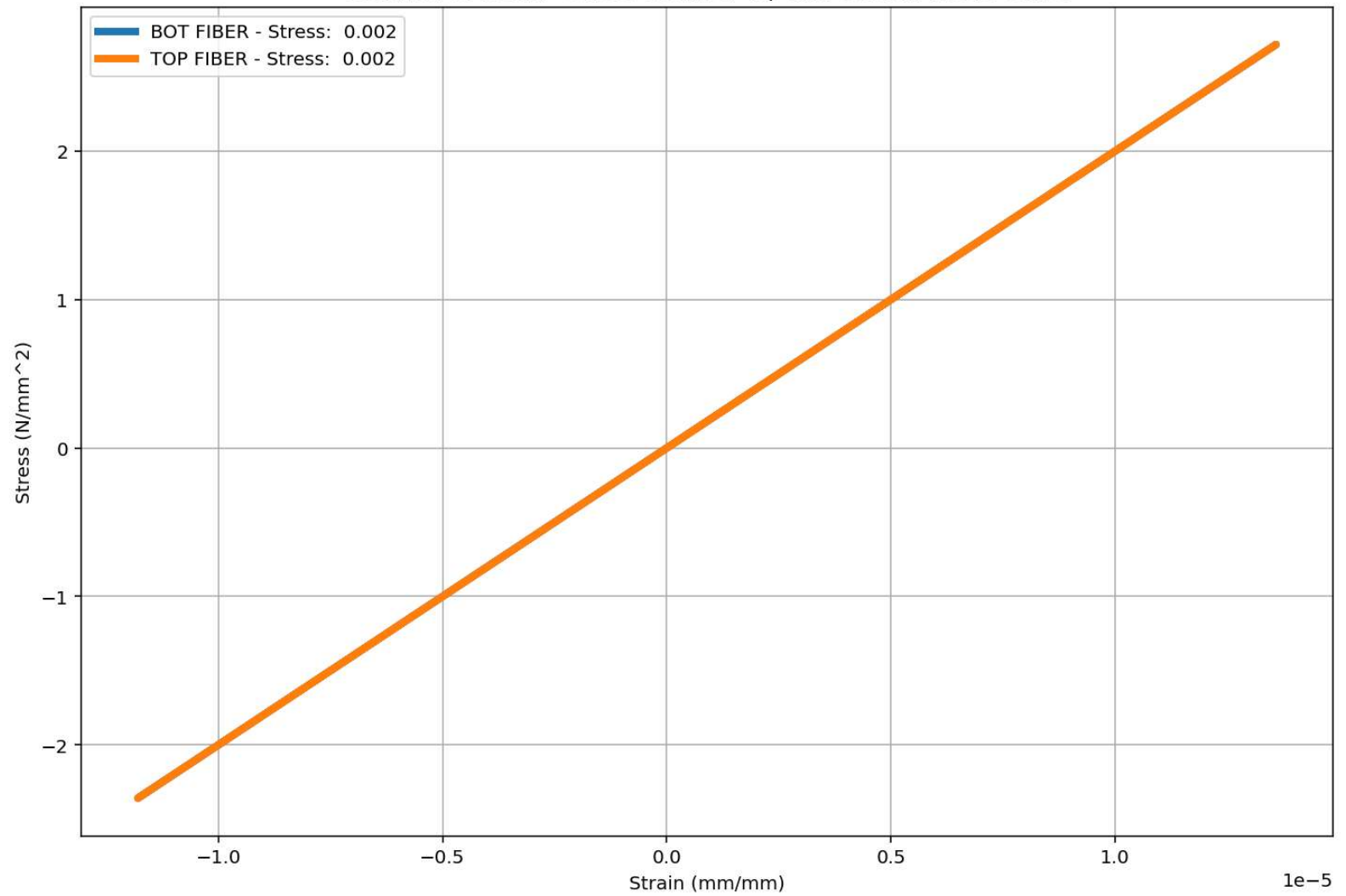


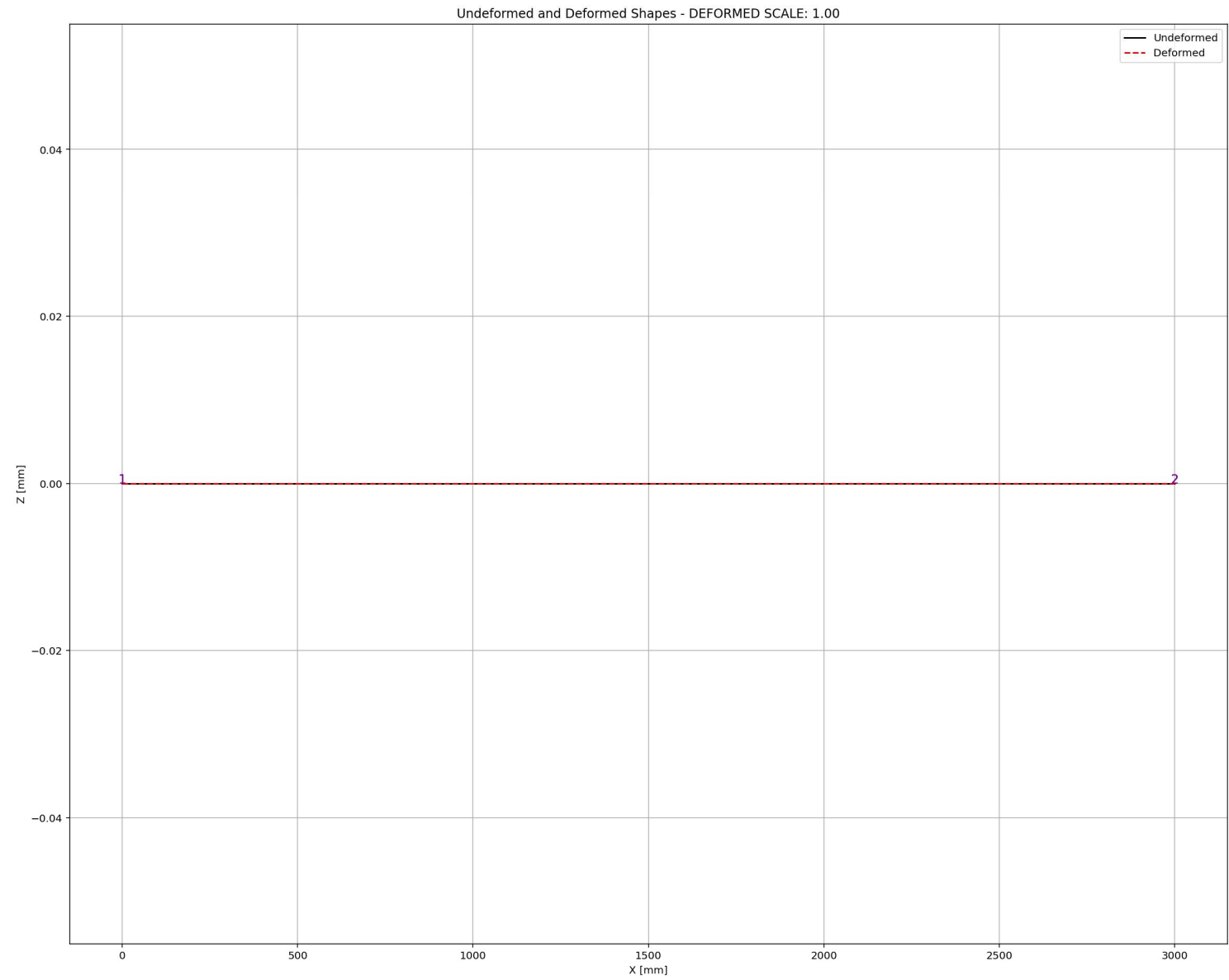
Node Displacements in Y Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis





Behavior of beam - Stress-strain of Top and Bottom Fibers Curve





FREE-VIBRATION ANALYSIS

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...EASE_M3_8_ANA\P(t)_BEAM_ONE_ELEMENT_RELEASE_M3.py

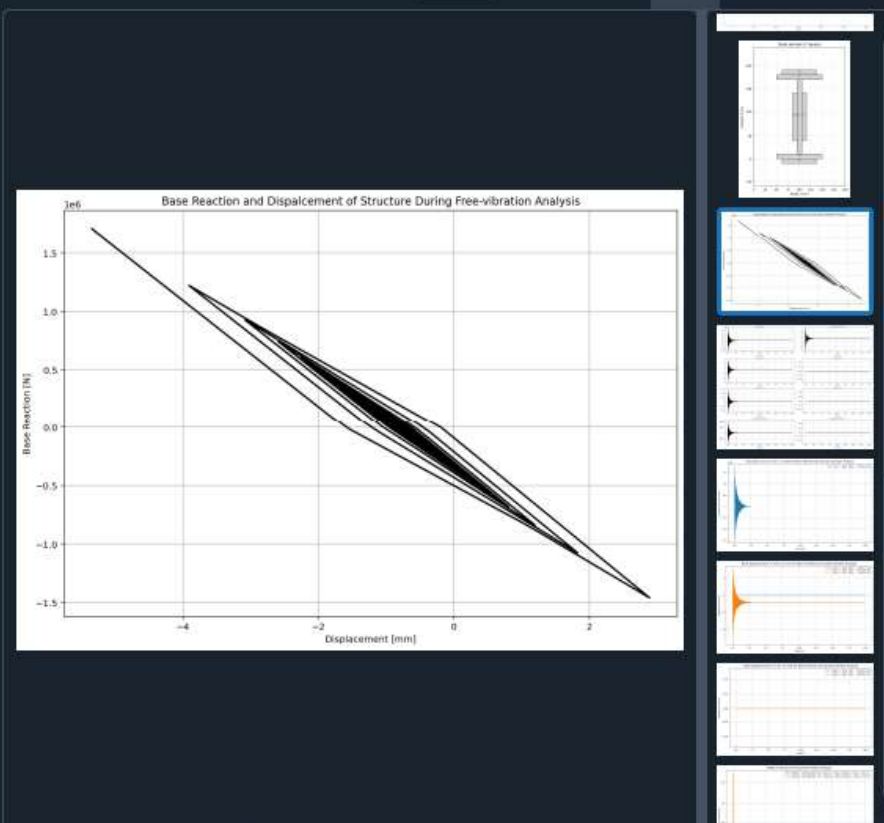
P(t)_BEAM_ONE_ELEMENT_RELEASE_M3.py X I_STEEL_PLATE_SECT...EIGHT_FUN_EXTRA.py X

```

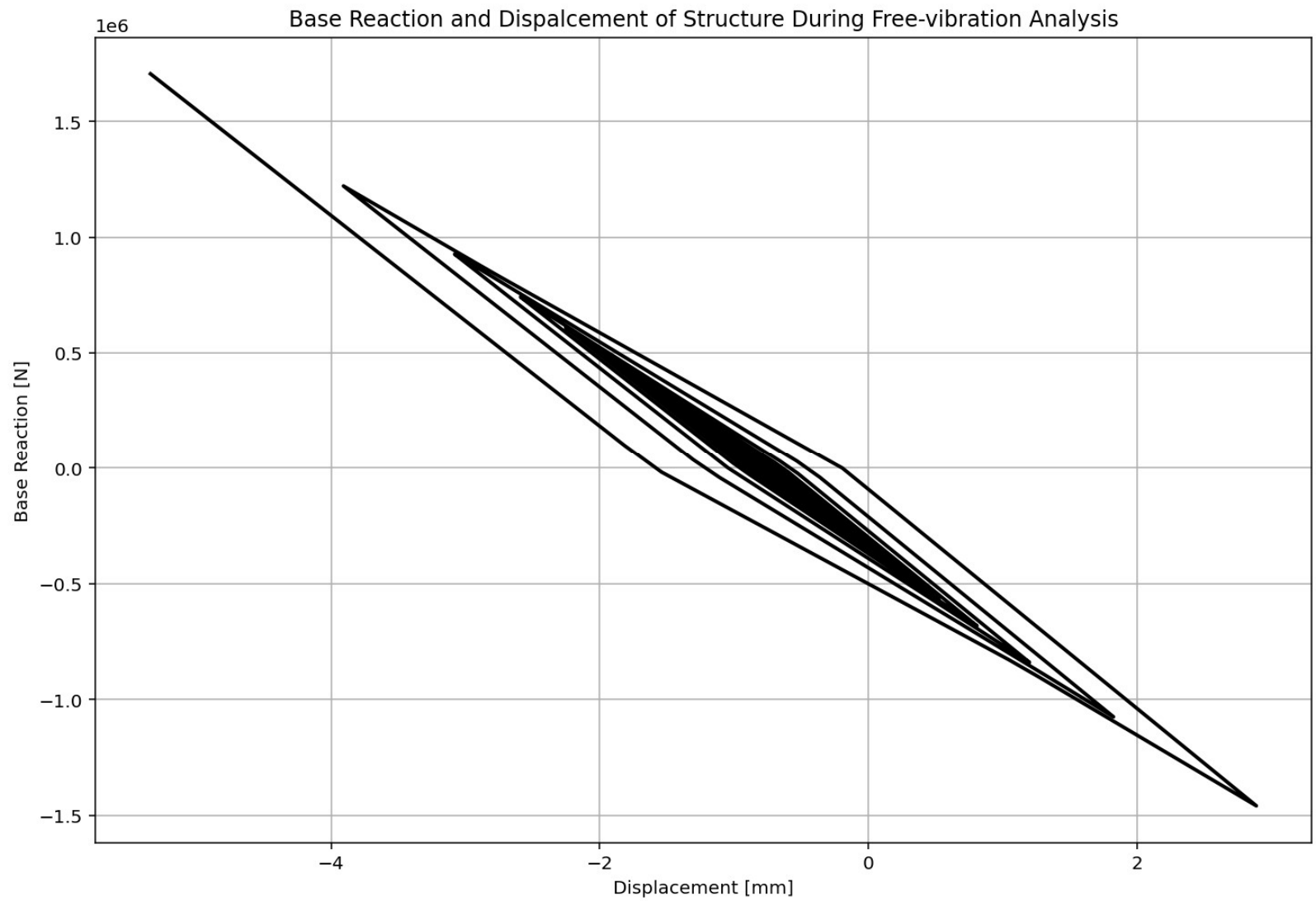
1294 # FREE-VIBRATION ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
1295 #ELE_TYPE = 'elasticBeamColumn' # MATERIAL LINEARITY
1296 ELE_TYPE = 'nonlinearBeamColumn' # MATERIAL AND GEOMETRIC NONLINEARITIES
1297 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1298 ANAL_TYPE = 'FREE-VIBRATION'
1299 DATA = BEAM_ONE_ELEMENT(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
1300
1301 (time_FV, reaction_FV, disp_mid_FV,
1302  ele_force_FV, ele_stress_FV, ele_strain_FV, ele_di_FV,
1303  node_displacementsX_FV, node_displacementsY_FV,
1304  dispX_FV, dispY_FV,
1305  veloX_FV, veloY_FV,
1306  accX_FV, accY_FV,
1307  stiffness_FV, PERIOD_FV, damping_ratio_FV,
1308  PERIOD_MIN_FV, PERIOD_MAX_FV,
1309  STRESSb_FV, STRAINb_FV, STRESSt_FV, STRAINt_FV, CURVATURE_FV) = DATA
1310
1311 XDATA = disp_mid_FV
1312 YDATA = reaction_FV
1313 XLABEL = 'Displacement [mm]'
1314 YLABEL = 'Base Reaction [N]'
1315 TITLE = 'Base Reaction and Dispalcement of Structure During Free-vibration Analysis'
1316 COLOR = 'black'
1317 SEMILOGY = False
1318 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1319
1320 PLOT_TIME_HISTORY(time_FV, reaction_FV, disp_mid_FV,
1321                   dispX_FV, dispY_FV,
1322                   veloX_FV, veloY_FV,
1323                   accX_FV, accY_FV)
1324
1325 # PLOT ELEMENTS AXIAL FORCE
1326 YLABEL = 'Element Axial Force [N]'
1327 TITLE = "elements force in X Dis vs Time for Beam Element During Free-vibration Analysis"

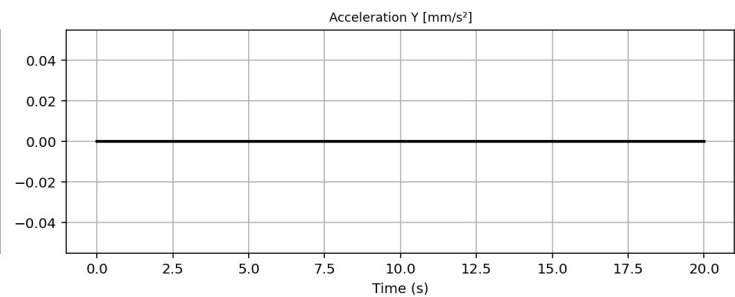
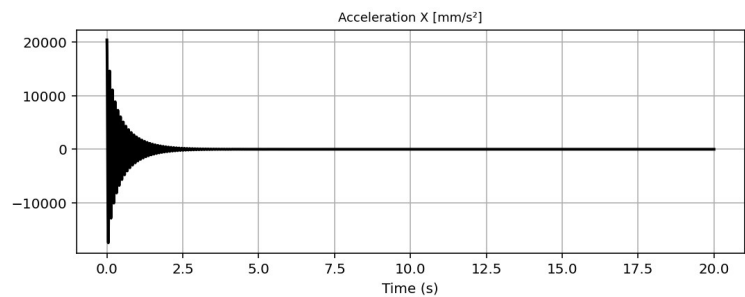
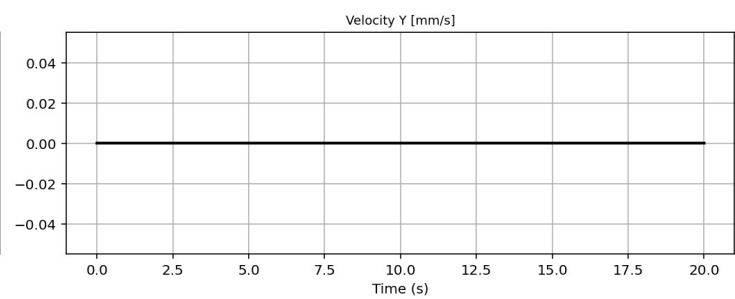
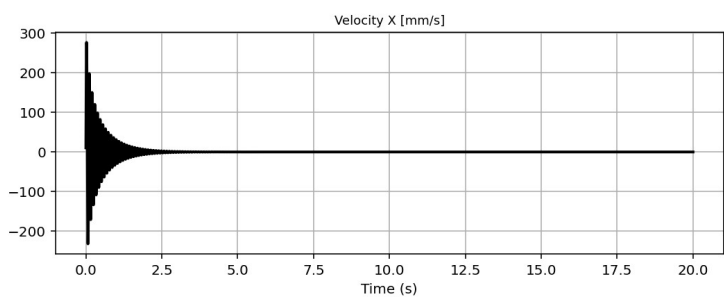
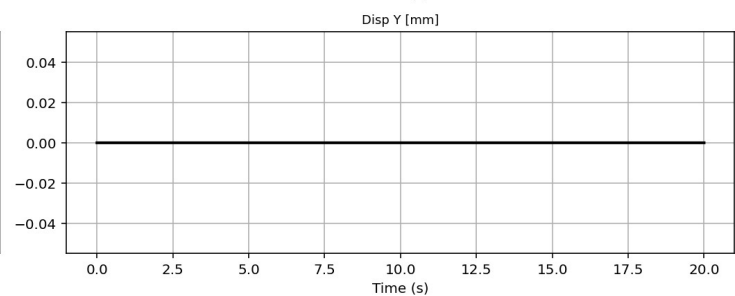
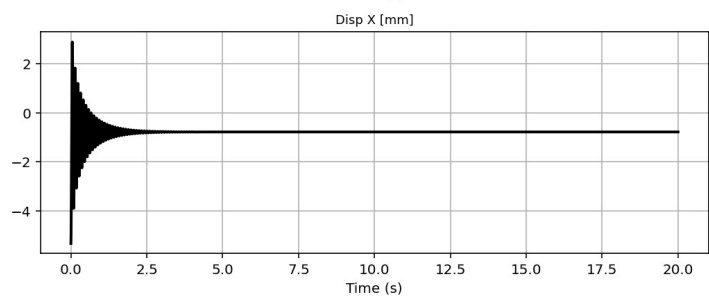
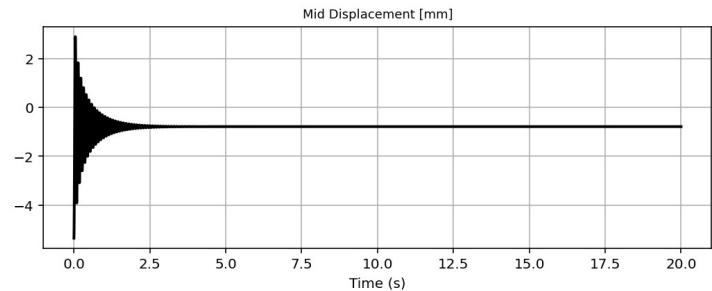
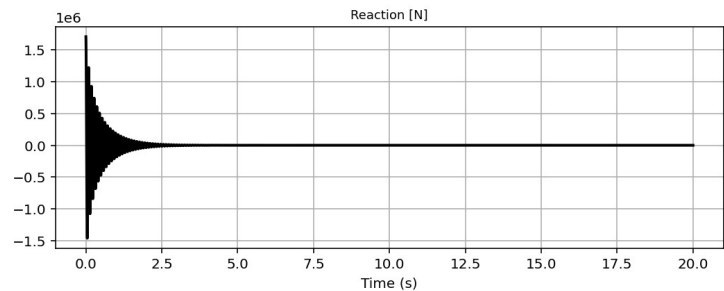
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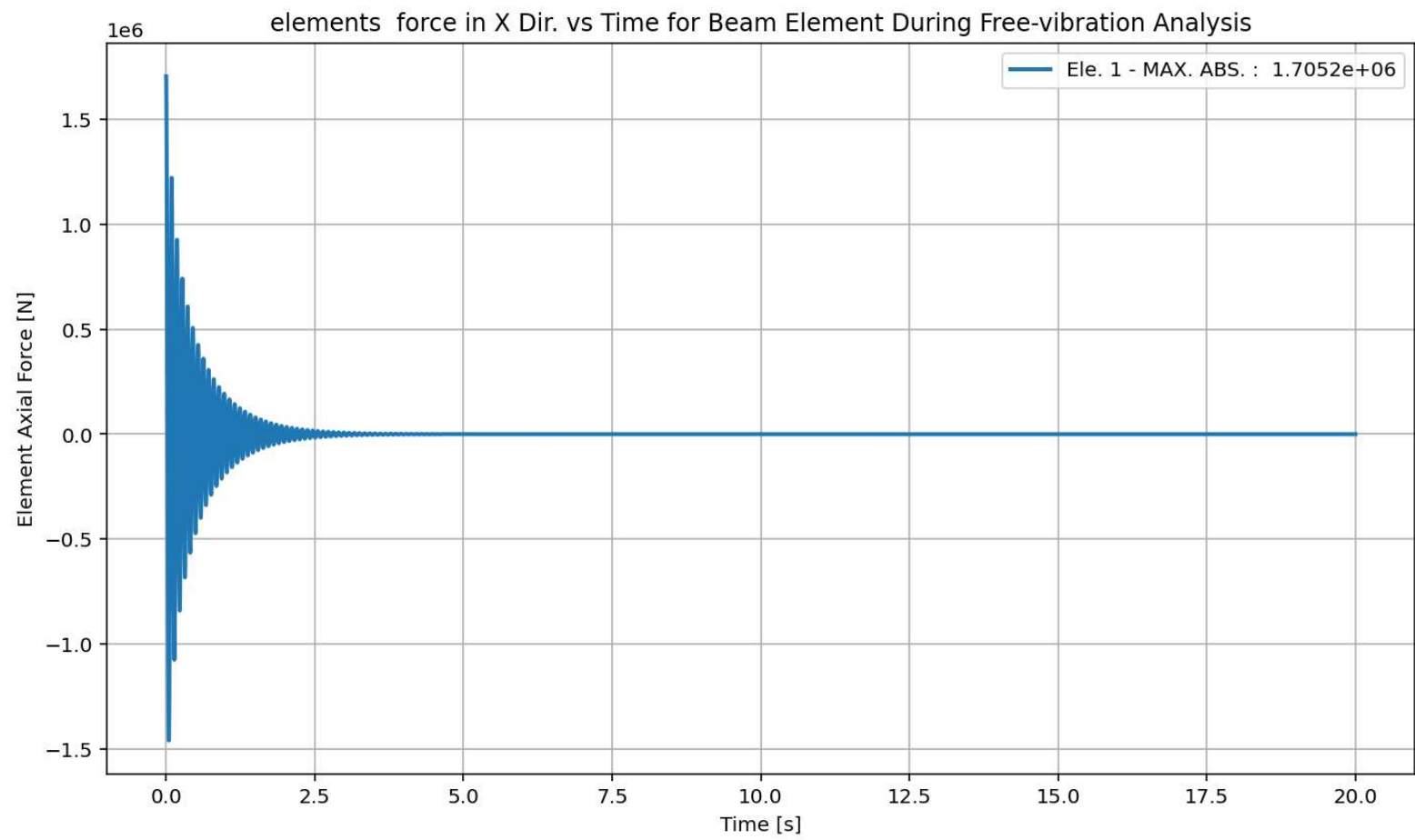
29 %

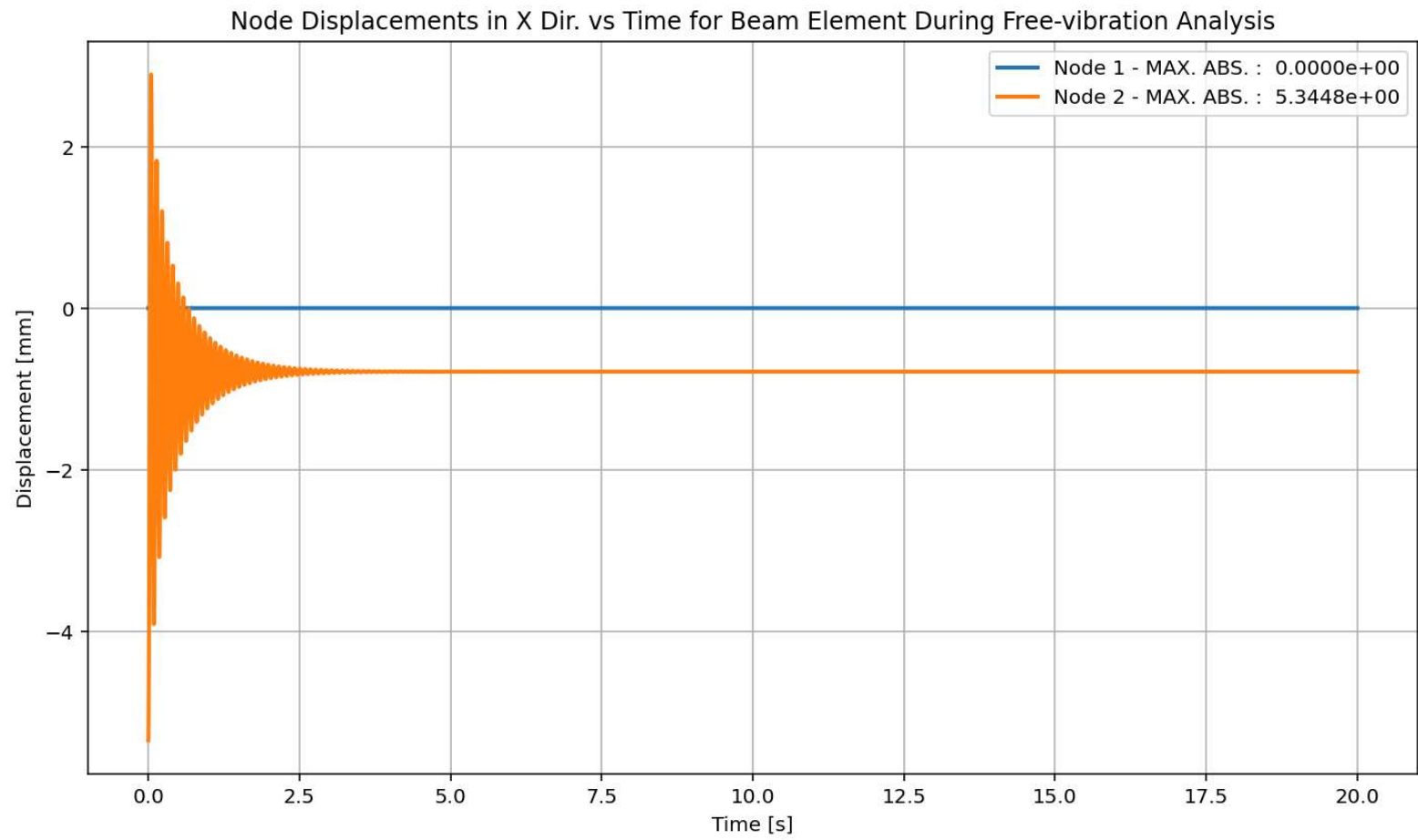


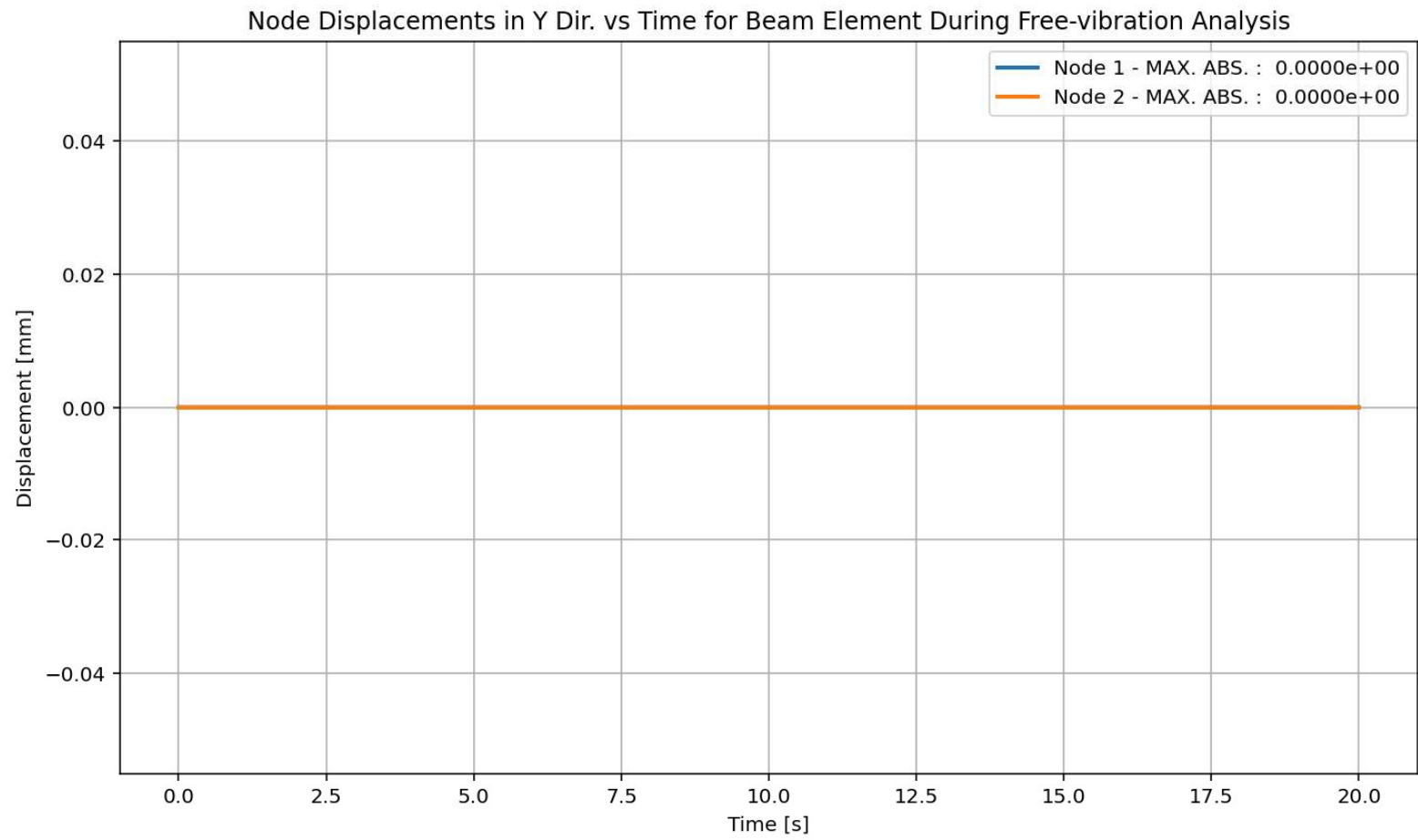
IPython Console Plots Variable Explorer Help Debugger History Files



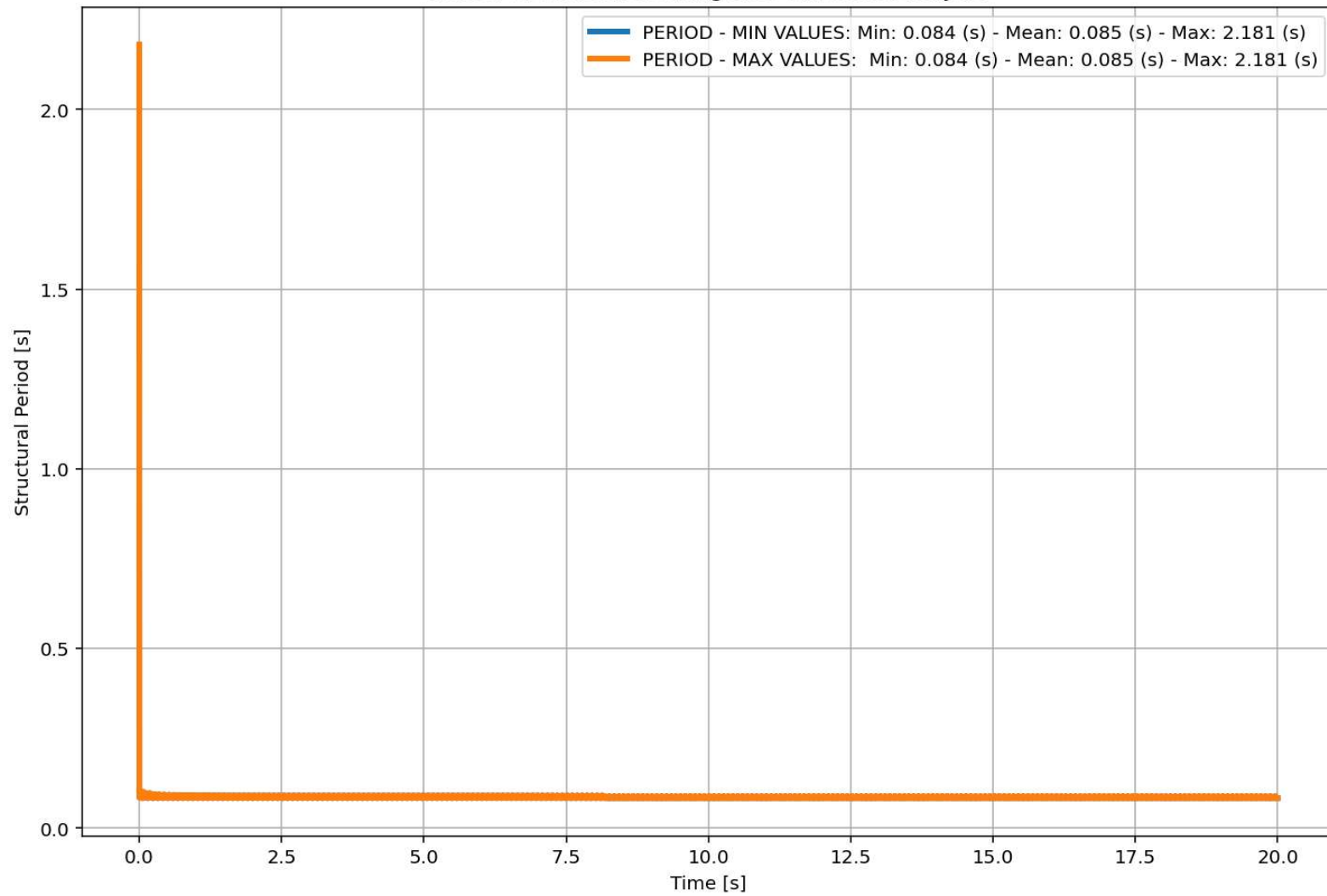


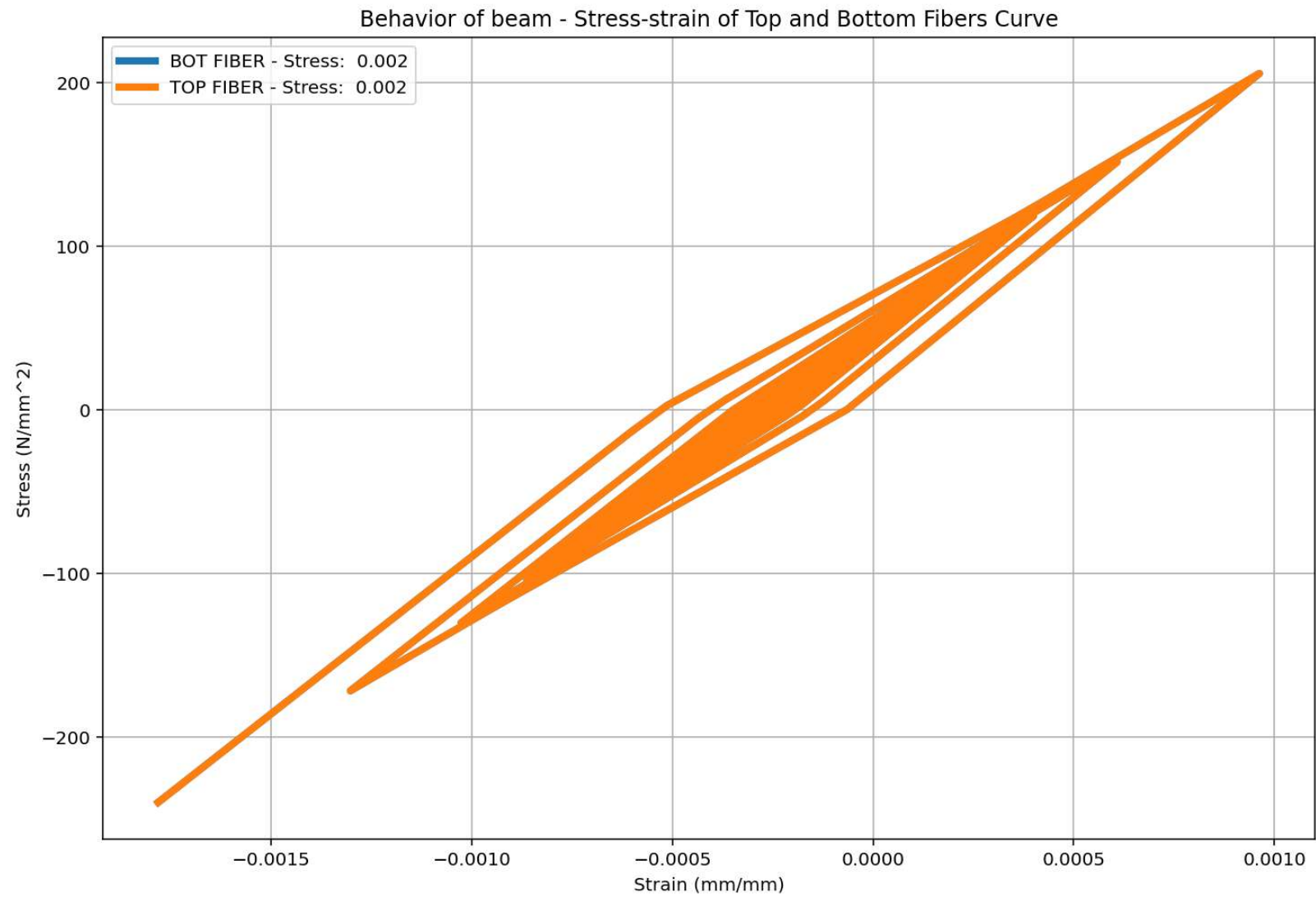


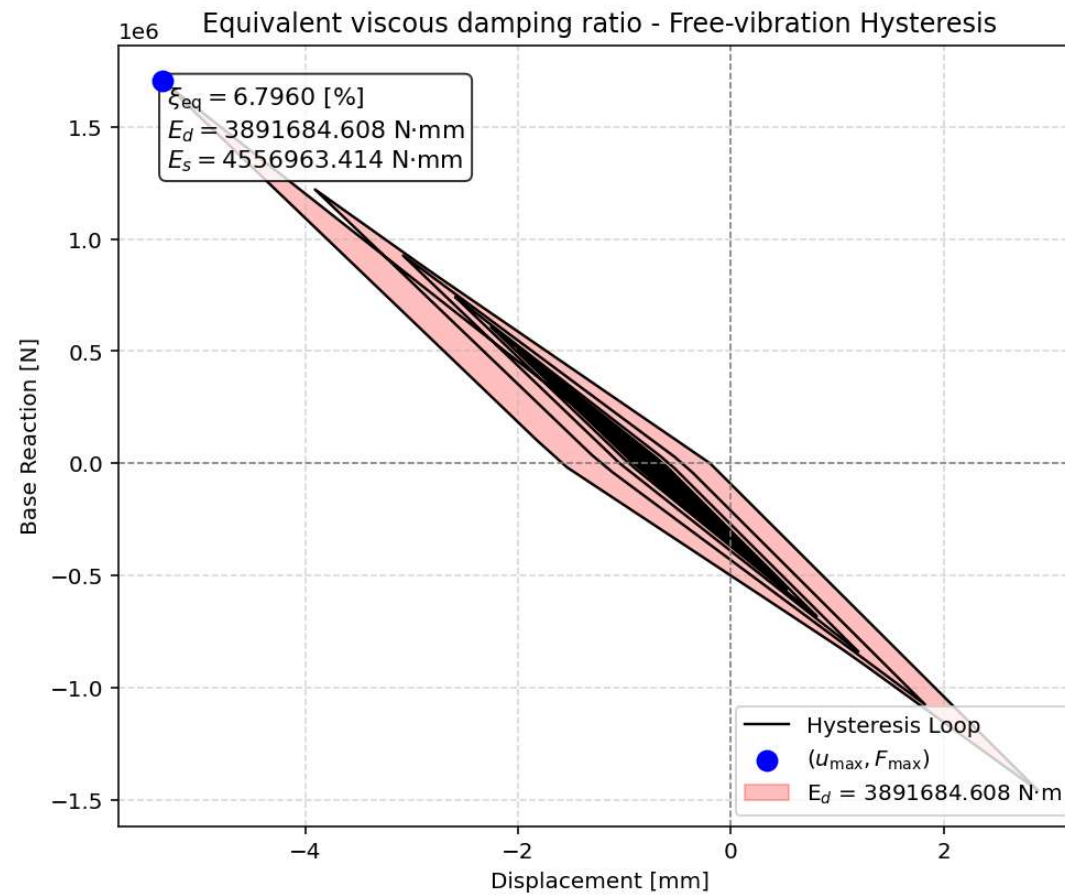


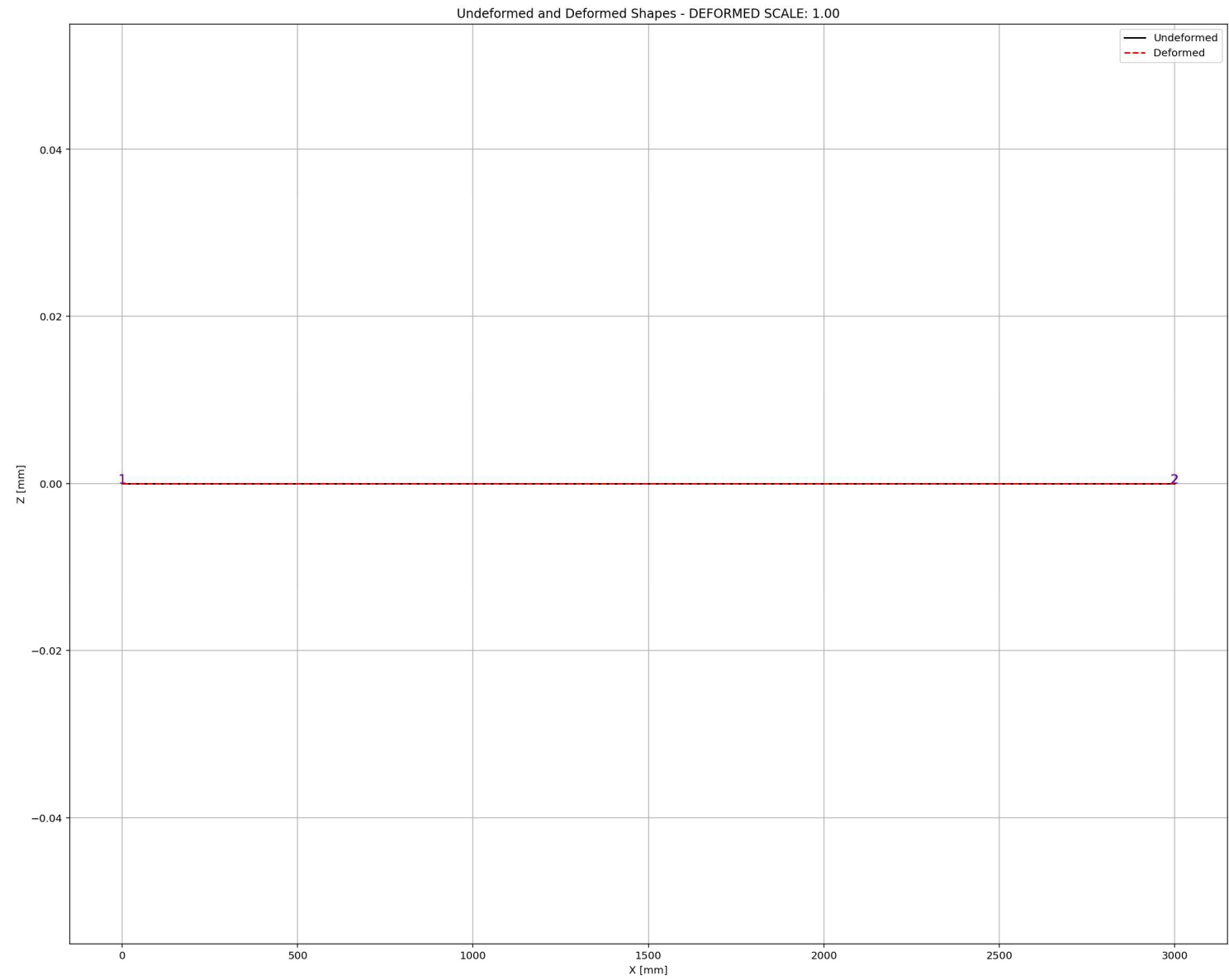


Period of Structure During Free-vibration Analysis







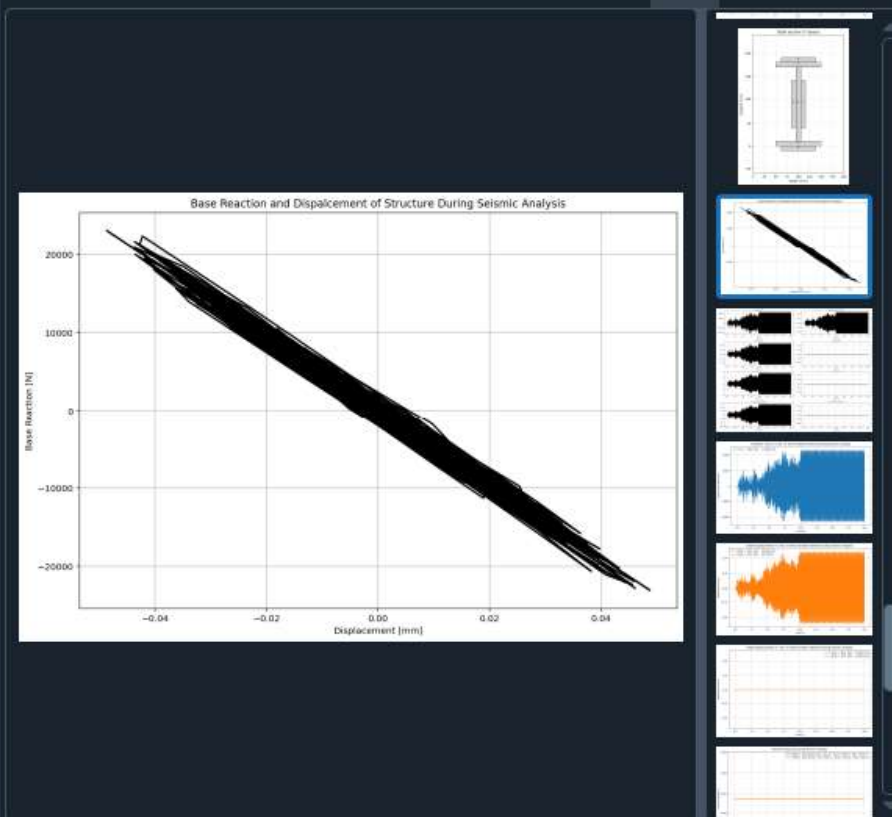


SEISMIC ANALYSIS

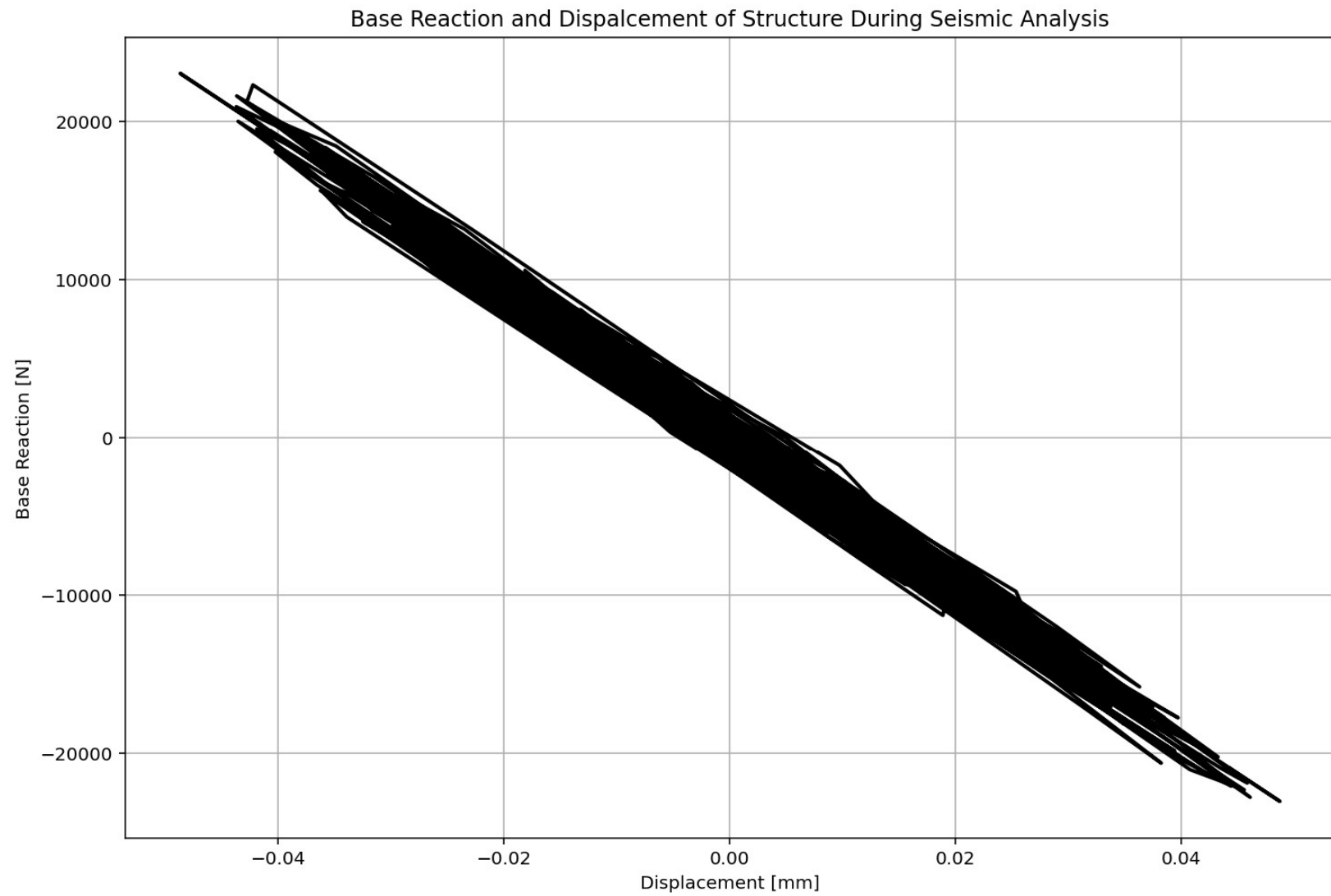
C:\Users\ DELL\Desktop\OPENSEES_FILES\+EIGHT_ANALY...EASE_M3_8_ANA\P(t)_BEAM_ONE_ELEMENT_RELEASE_M3.py

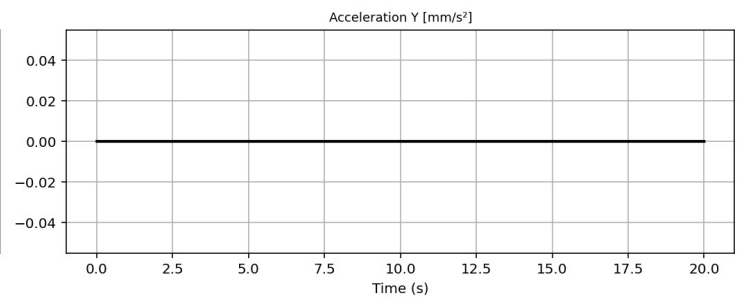
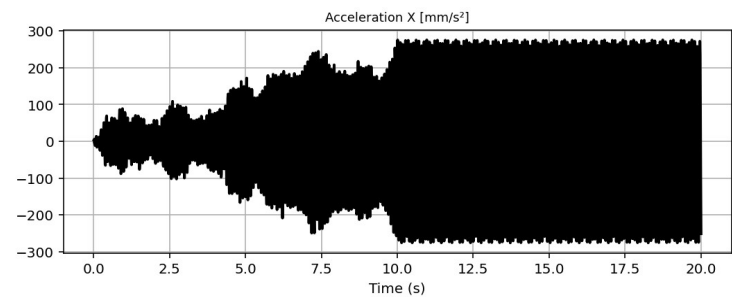
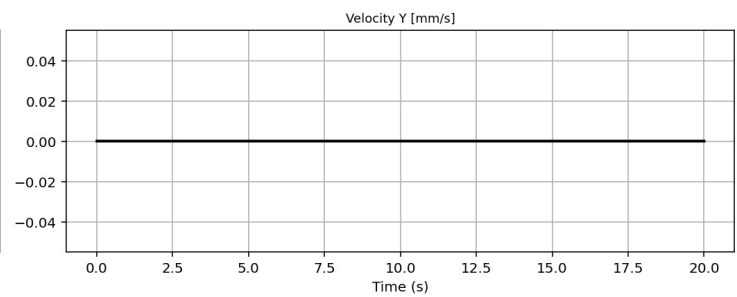
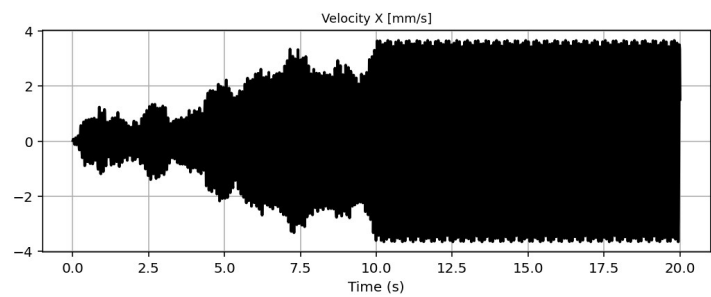
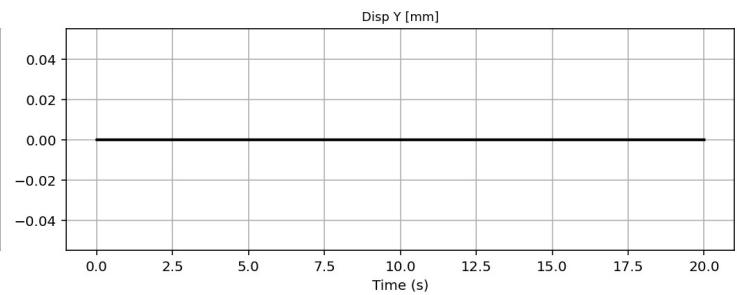
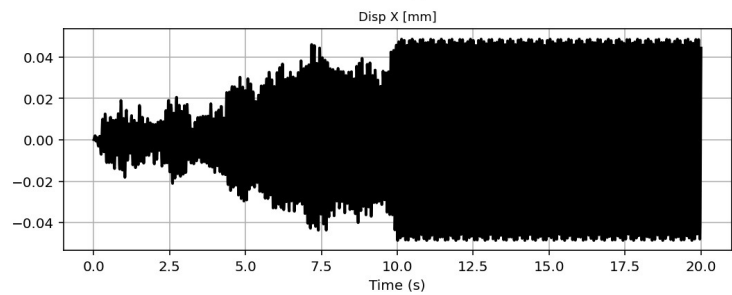
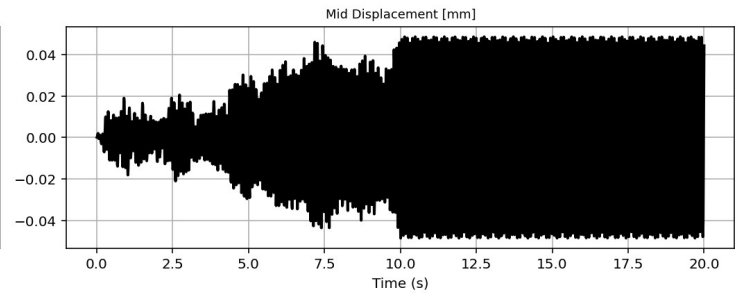
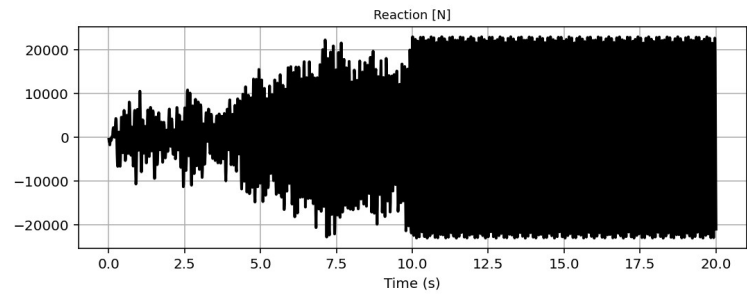
```
P(t)_BEAM_ONE_ELEMENT_RELEASE_M3.py X I_STEEL_PLATE_SECT...EIGHT_FUN_EXTRA.py X
1380 # SEISMIC ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
1381 #ELE_TYPE = 'elasticBeamColumn' # MATERIAL LINEARITY
1382 ELE_TYPE = 'nonlinearBeamColumn' # MATERIAL AND GEOMETRIC NONLINEARITIES
1383 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1384 ANAL_TYPE = 'SEISMIC'
1385
1386 DATA = BEAM_ONE_ELEMENT(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
1387
1388 (time_SEI, reaction_SEI, disp_mid_SEI,
1389  ele_axialforce_SEI, ele_stress_SEI, ele_strain_SEI, ele_di_SEI,
1390  node_displacementsX_SEI, node_displacementsY_SEI,
1391  dispX_SEI, dispY_SEI,
1392  veloX_SEI, veloY_SEI,
1393  accX_SEI, accY_SEI,
1394  stiffness_SEI, PERIOD_SEI, damping_ratio_SEI,
1395  PERIOD_MIN_SEI, PERIOD_MAX_SEI,
1396  STRESSb_SEI, STRAINb_SEI, STRESSt_SEI, STRAINT_SEI, CURVATURE_SEI) = DATA
1397
1398
1399 XDATA = disp_mid_SEI
1400 YDATA = reaction_SEI
1401 XLABEL = 'Displacement [mm]'
1402 YLABEL = 'Base Reaction [N]'
1403 TITLE = 'Base Reaction and Dispalcement of Structure During Seismic Analysis'
1404 COLOR = 'black'
1405 SEMIOLOGY = False
1406 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMIOLOGY)
1407
1408 PLOT_TIME_HISTORY(time_SEI, reaction_SEI, disp_mid_SEI,
1409                   dispX_SEI, dispY_SEI,
1410                   veloX_SEI, veloY_SEI,
1411                   accX_SEI, accY_SEI)
1412
1413 # PLOT ELEMENTS AXIAL FORCE
```

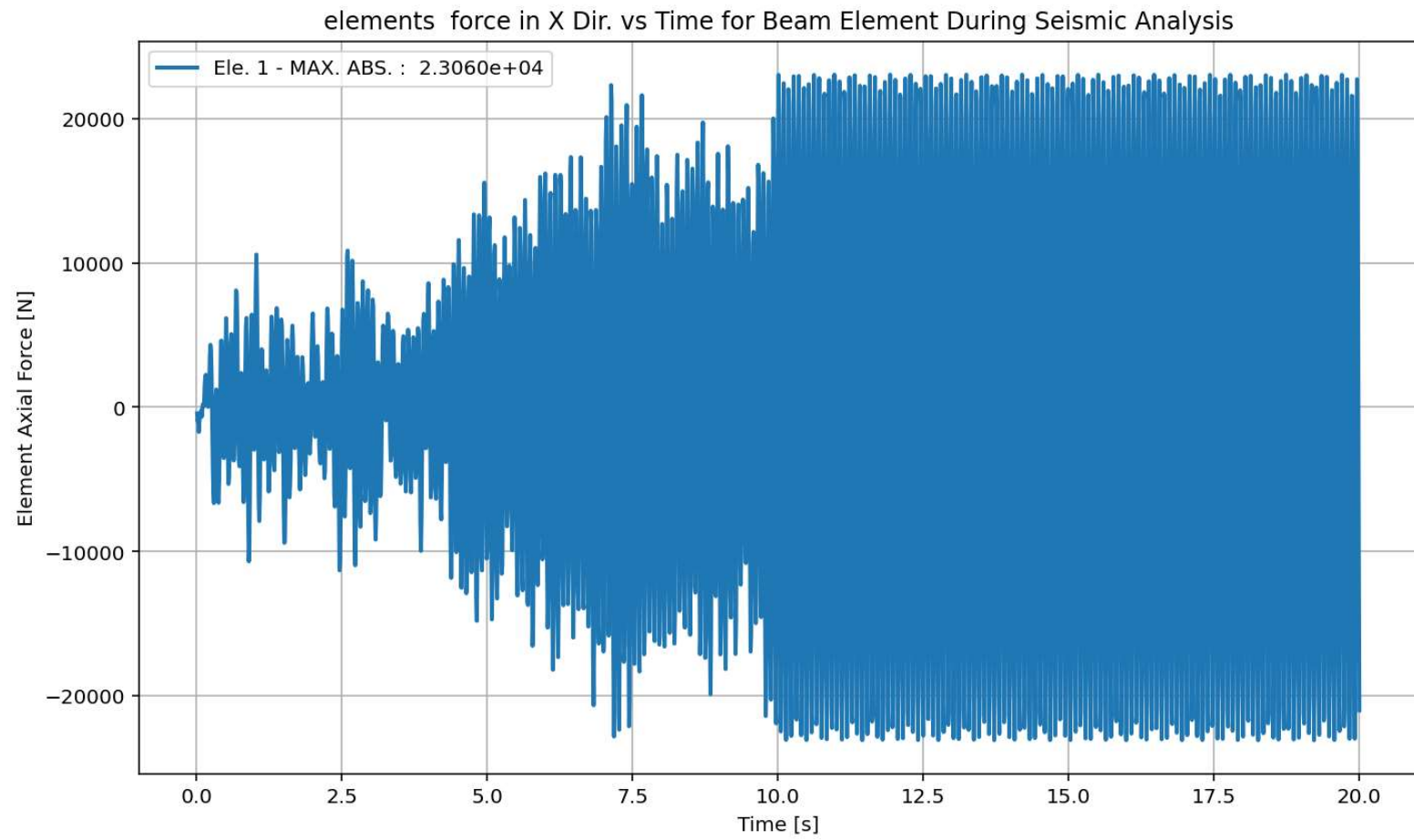
...ATION_CAPACITY_INDEX\02_BEAM_ONE_ELEMENT_RELEASE_M3_8_ANA 29 %

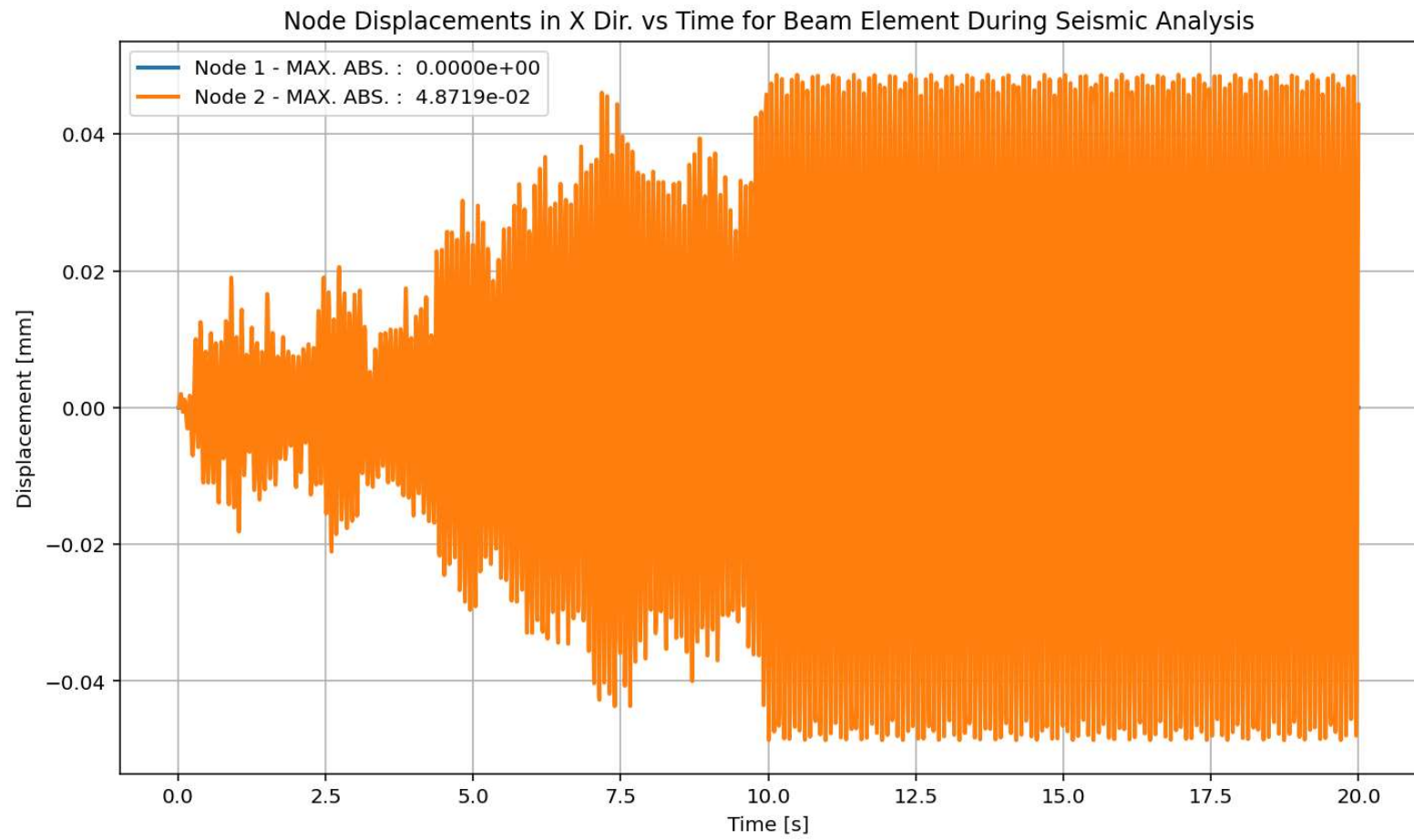


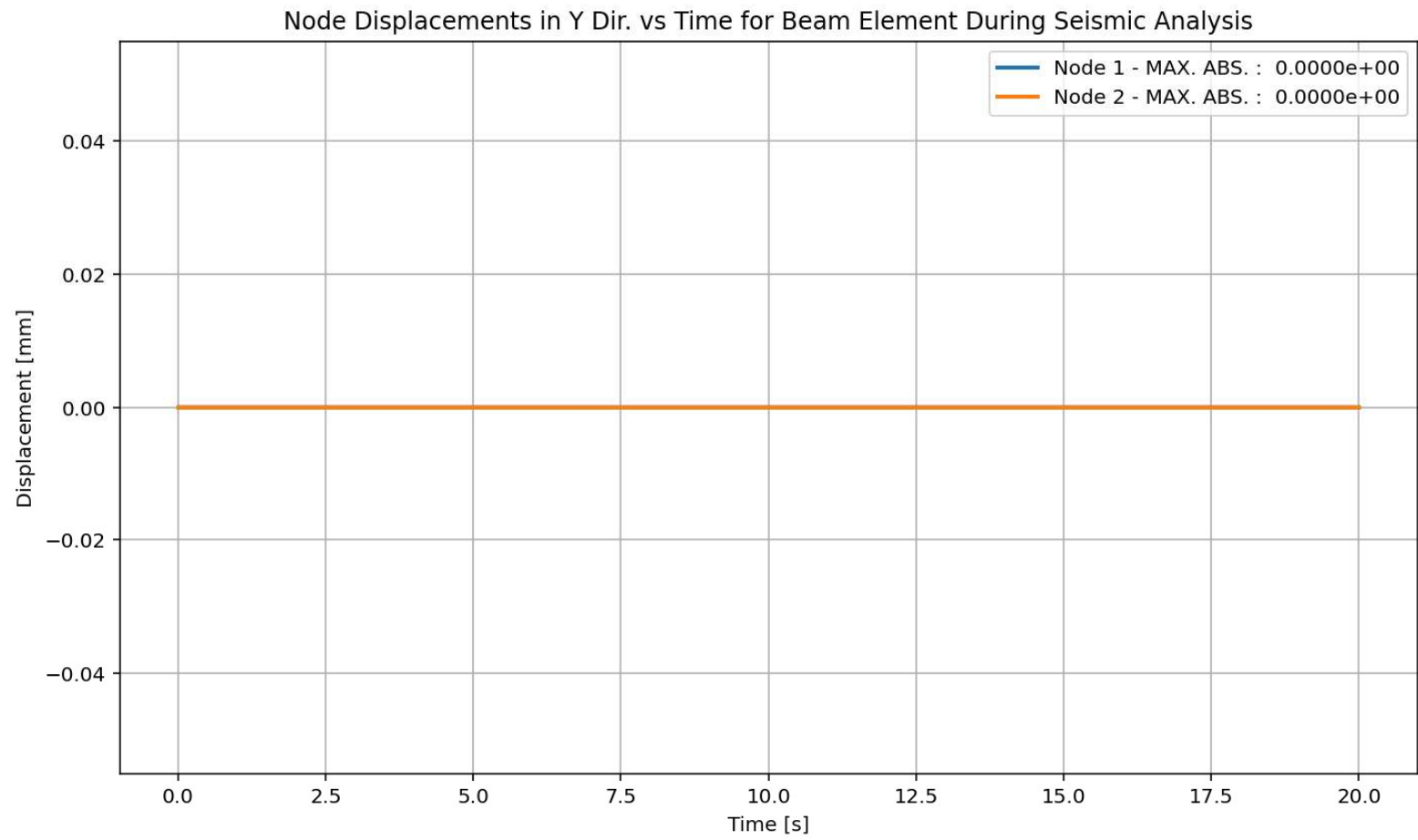
IPython Console Plots Variable Explorer Help Debugger History Files

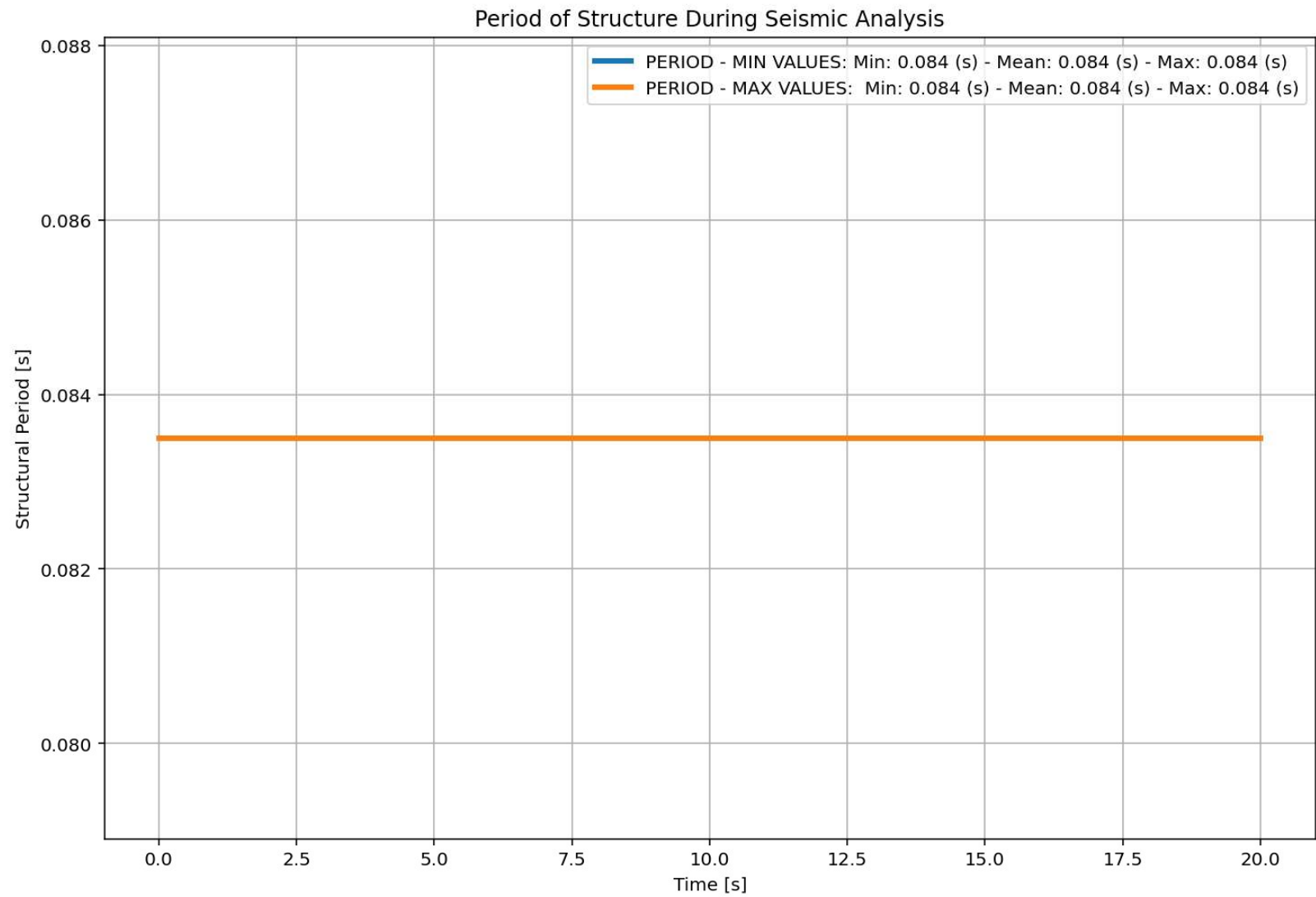




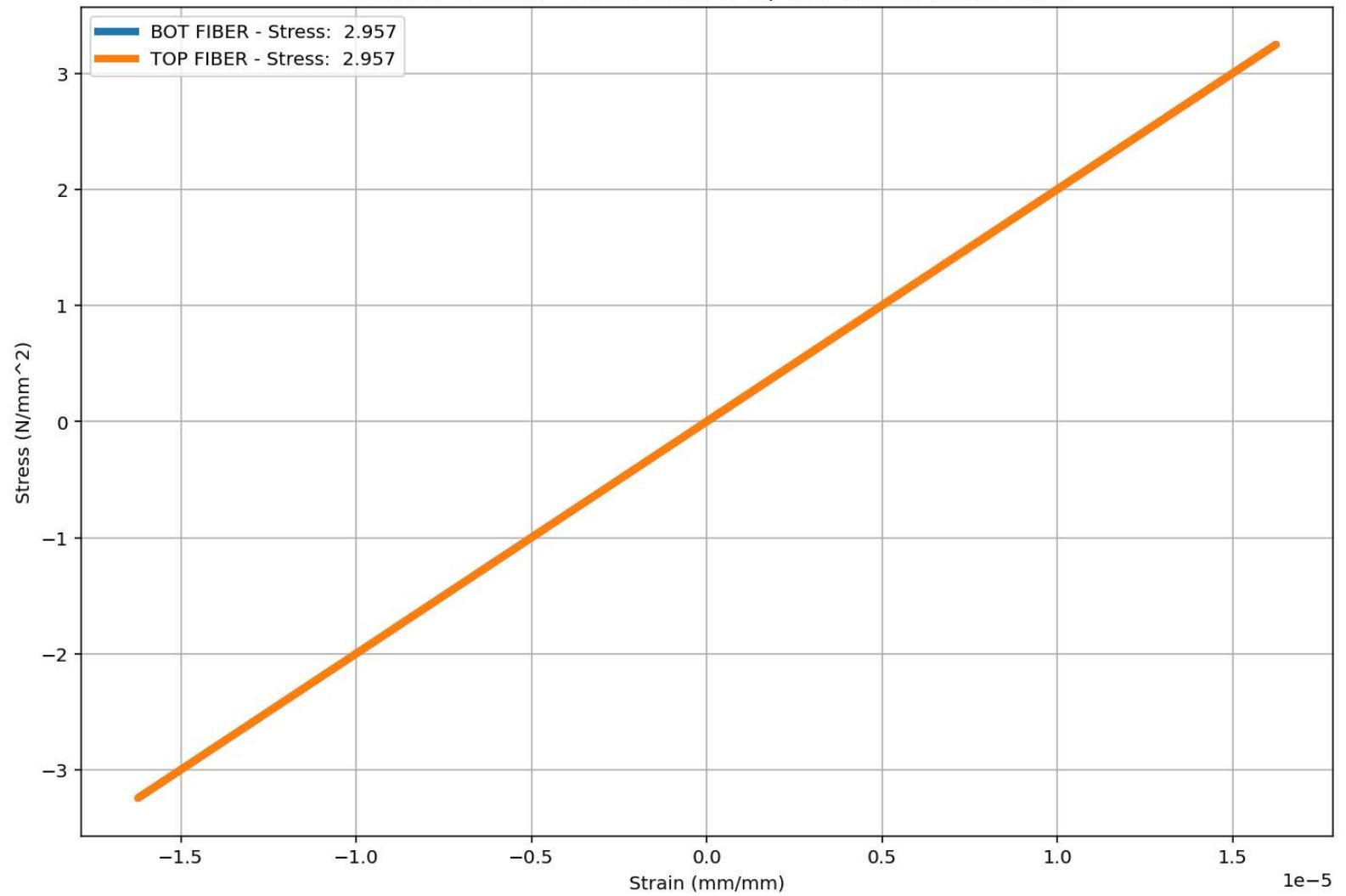


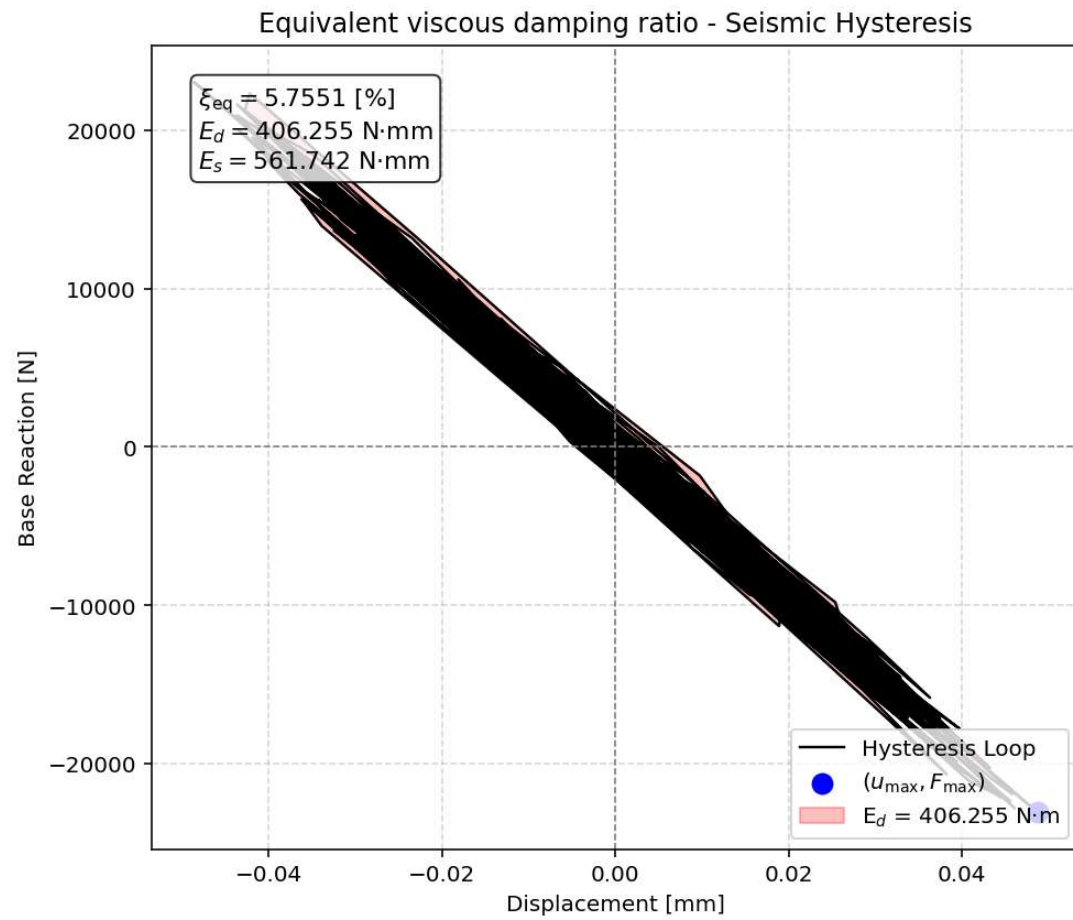


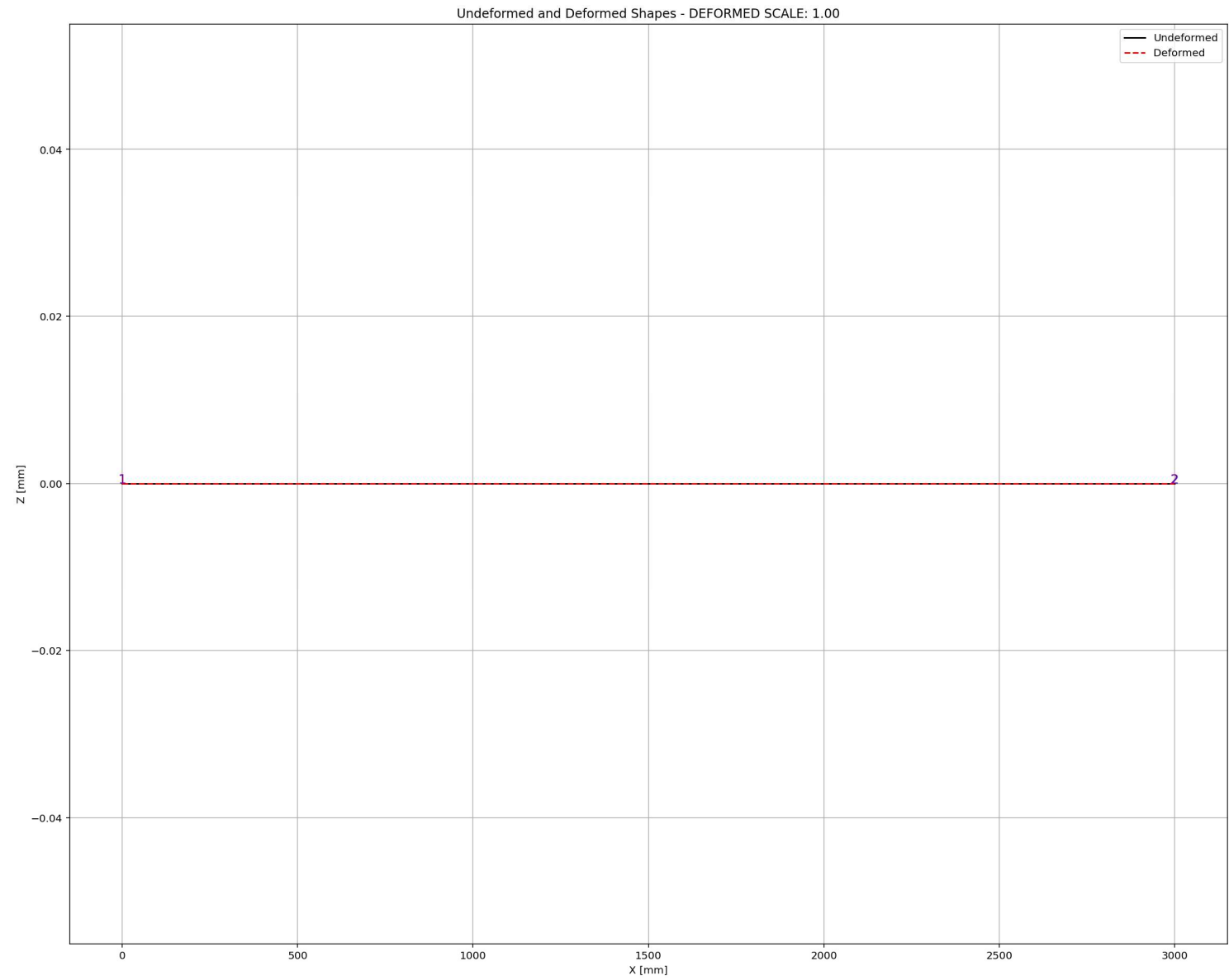


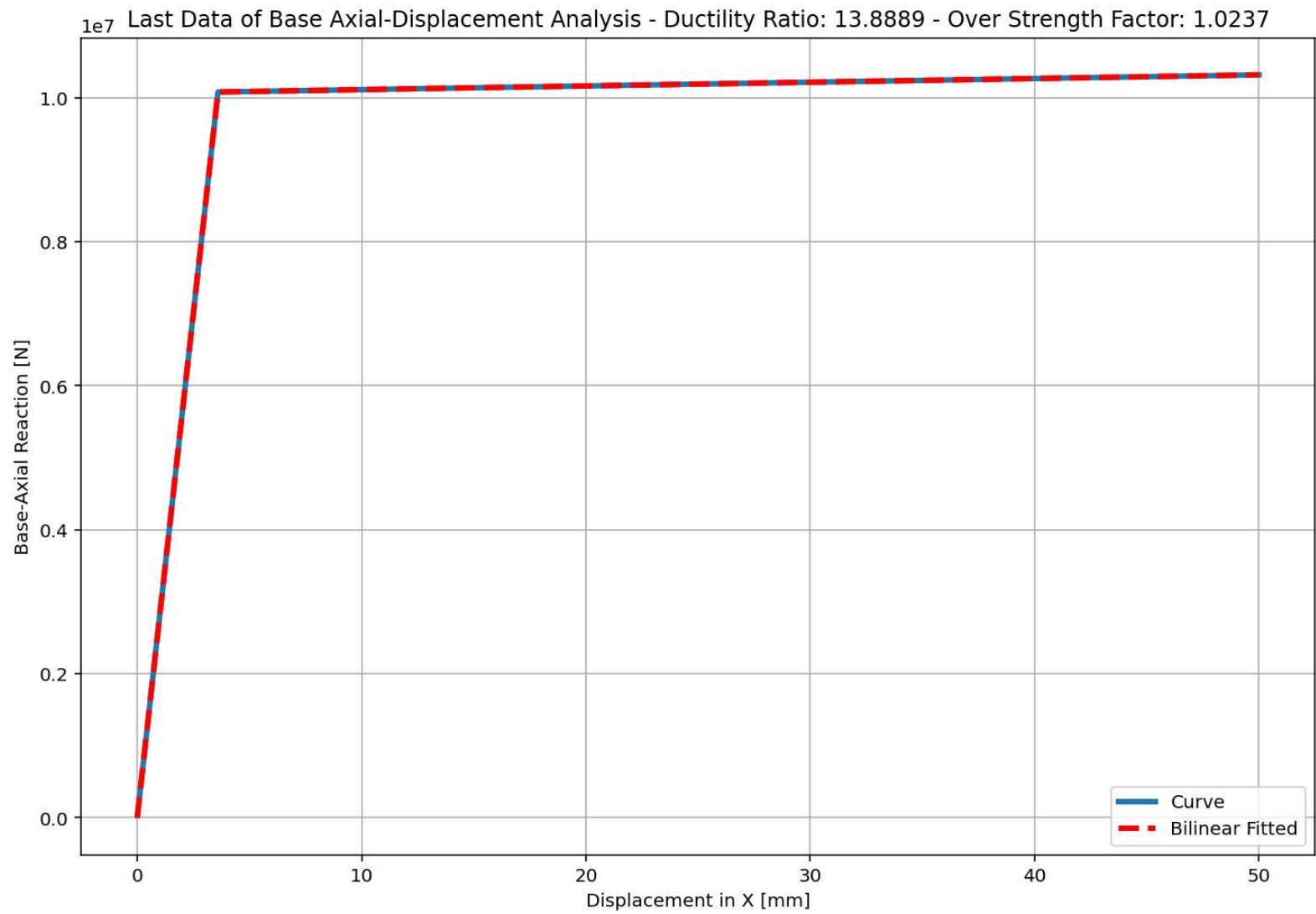


Behavior of beam - Stress-strain of Top and Bottom Fibers Curve









+=====+

= Analysis curve fitted =

Disp Base Shear

[[0.00000000e+00 0.00000000e+00]

[3.59999784e+00 1.00799940e+07]

[5.00000000e+01 1.03185804e+07]]

+=====+

+-----+

Structure Elastic Stiffness : 2800000.00

Structure Plastic Stiffness : 206371.61

Structure Tangent Stiffness : 5141.95

Structure Ductility Ratio : 13.89

Structure Over Strength Factor: 1.02

+-----+

Over Strength Coefficient (Ω_0): 1.0237

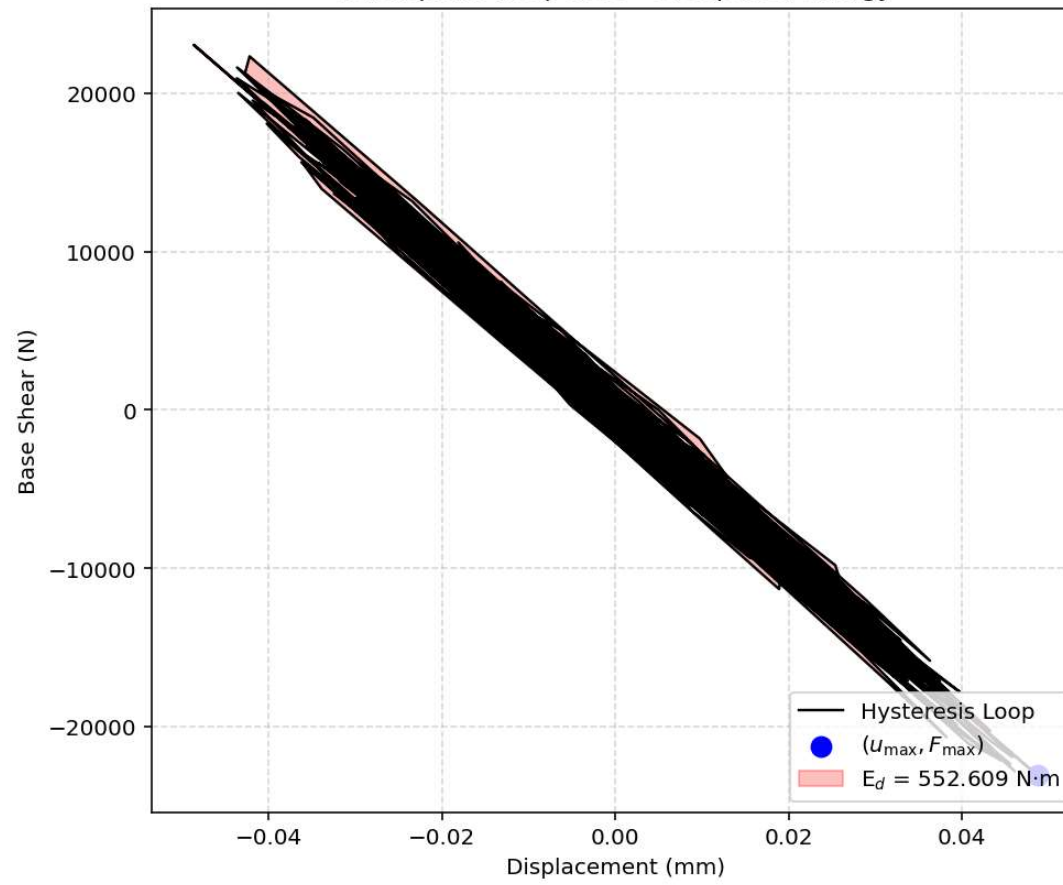
Displacement Ductility Ratio (μ): 13.8889

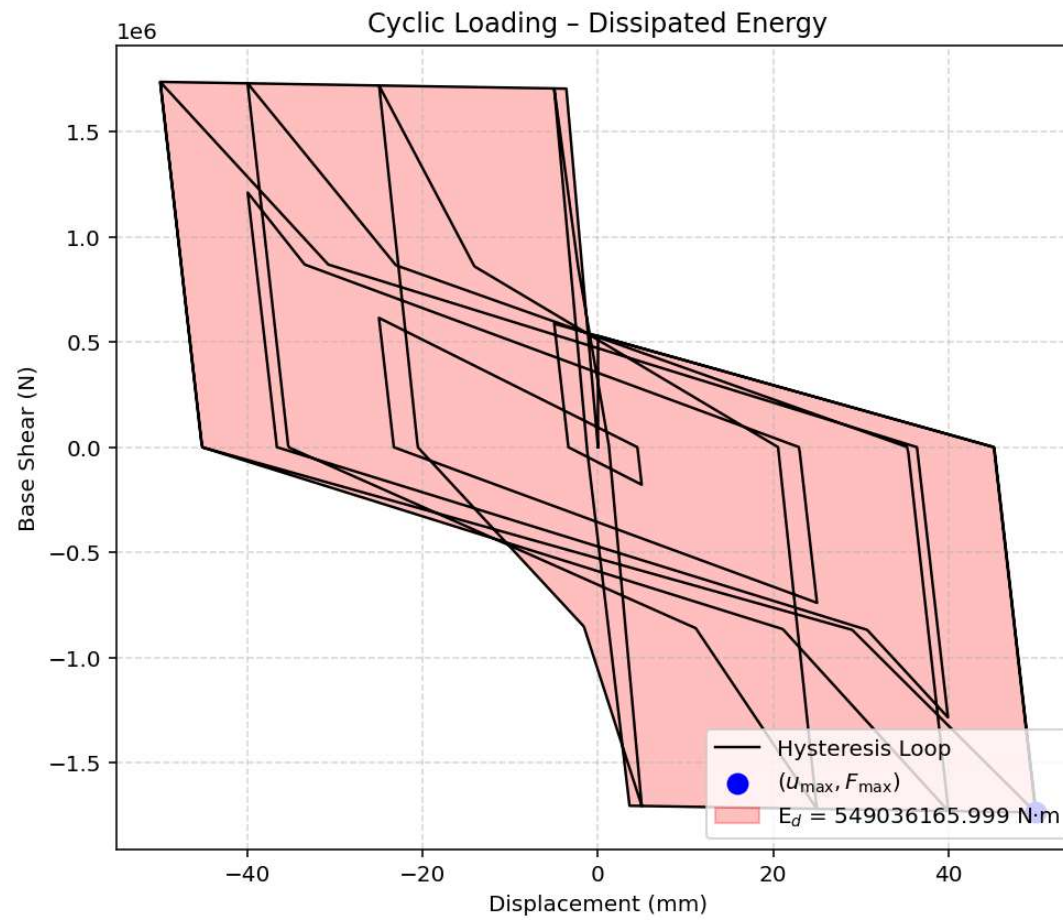
Ductility Coefficient (R_μ): 13.8889

Structural Behavior Coefficient (R): 14.2176

Structural Ductility Damage Index in X Direction: -7.6537 (%)

Earthquake Response - Dissipated Energy





Dissipated Energy from Earthquake= 552.61 N·mm

Dissipated Energy from Cyclic Displacement= 549036166.00 N·mm

DISSIPATED ENERGY CAPACITY INDEX: 0.000 [%]

ZONE 1: VERY MINOR DAMAGE

